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SENIOR LIVING CARE COORDINATION PLATFORMS

Abstract

Provided herein is a care coordination support platform (“CCSP”) computer system including at least one processor in communication with at least one memory device for coordinating care. The processor is programmed to: (i) register a user through an application, wherein the user inputs personal and scheduling data into the application, (ii) register a caregiver associated with the user through the application, wherein the caregiver inputs personal and scheduling data into the application, (iii) receive input from at least one of the user and the caregiver defining at least one event associated with the user, (iv) assign the at least one event to at least one caregiver based upon the personal and scheduling data of the caregivers, (v) create a care schedule of the user, wherein the care schedule of the user includes the at least one assigned event, and (vi) display the care schedule to the at least one caregiver.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application claims priority to U.S. Provisional Patent Application Ser. No. 62/870,515, filed Jul. 3, 2019, entitled “SENIOR LIVING CARE COORDINATION PLATFORMS,” U.S. Provisional Patent Application Ser. No. 62/872,014, filed Jul. 9, 2019, entitled “SENIOR LIVING CARE COORDINATION PLATFORMS,” and U.S. Provisional Patent Application Ser. No. 62/874,258, filed Jul. 15, 2019, entitled “SENIOR LIVING CARE COORDINATION PLATFORMS,” the entire contents and disclosures of which are hereby incorporated herein by reference in their entirety.

FIELD OF THE INVENTION

[0002] The present disclosure relates to senior living computer platforms and, more particularly, to systems and methods for using a senior living computer platform to coordinate a care schedule for a senior user, wherein the care schedule includes schedules of caregivers and other service providers for the senior user.

BACKGROUND

[0003] At least some conventional computer networks (e.g., matching platforms) have enabled caregivers (e.g., family members, friends, and care service providers) associated with senior users to coordinate care for the senior user. However, conventional systems usually merely keep a schedule of the coordinated care, and may not provide additional functionality. Further, in the conventional systems, caregivers may have to coordinate the care between the other caregivers before the caregivers can prepare a schedule. Known systems may have other drawbacks as well.

BRIEF SUMMARY

[0004] The present embodiments may relate to systems and methods for electronically coordinating and maintaining a care schedule of a senior user for caregivers associated with helping the senior user. The system may include a care coordination support computing device, one or more client devices, one or more chatbot servers, one or more sensor servers, and/or one or more databases.

[0005] In one aspect, a care coordination support platform computer system including at least one processor in communication with at least one memory device for coordinating care may be provided. The processor may be programmed to: (i) register a user through an application, wherein the user inputs personal and scheduling data into the application, (ii) register a caregiver associated

with the user through the application, wherein the caregiver inputs personal and scheduling data into the application, and wherein the caregiver is one caregiver out of a plurality of caregivers associated with the user, (iii) receive input from at least one of the user and the caregiver defining at least one event associated with the user, wherein the event includes at least one of a task, an activity, and an appointment, (iv) assign the at least one event to at least one caregiver of the plurality of caregivers based upon the personal and scheduling data of the plurality of caregivers, (v) create a care schedule of the user, wherein the care schedule of the user includes the at least one assigned event, and/or (vi) display, via the application, the care schedule to the at least one caregiver on at least one device associated with the at least one caregiver. The care coordination support platform computer system may include additional, less, or alternate functionality, including that discussed elsewhere herein.

[0006] In another aspect, a computer-implemented method for coordinating care may be provided, and the computer-implemented method may be performed by a care coordination support platform computer system that includes at least one processor in communication with at least one memory device. The computer-implemented method may include: (i) registering a user through an application, wherein the user inputs personal and scheduling data into the application, (ii) registering a caregiver associated with the user through the application, wherein the caregiver inputs personal and scheduling data into the application, and wherein the caregiver is one caregiver out of a plurality of caregivers associated with the user, (iii) receiving input from at least one of the user and the caregiver defining at least one event associated with the user, wherein the event includes at least one of a task, an activity, and an appointment, (iv) assign the at least one event to at least one caregiver of the plurality of caregivers based upon the personal and scheduling data of the plurality of caregivers, (v) creating a care schedule of the user, wherein the care schedule of the user includes the at least one assigned event, and/or (vi) displaying, via the application, the care schedule to the at least one caregiver on at least one device associated with the at least one caregiver. The method may include additional, less, or alternate actions, including those discussed elsewhere herein.

[0007] In yet another aspect, a non-transitory computer-readable media having computer-executable instructions embodied thereon may be provided. When executed by a care coordination support platform including a processor in communication with at least one memory device, the computer executable instructions may cause the processor to: (i) register a user through an application, wherein the user inputs personal and scheduling data into the application, (ii) register a caregiver associated with the user through the application, wherein the caregiver inputs personal and scheduling data into the application, and wherein the caregiver is one caregiver out of a plurality of caregivers associated with the user, (iii) receive input from at least one of the user and the caregiver defining at least one event associated with the user, wherein the event includes at least one of a task, an activity, and an appointment, (iv) assign the at least one event to at least one caregiver of the plurality of caregivers based upon the personal and scheduling data of the plurality of caregivers, (v) create a care schedule of the user, wherein the care schedule of the user includes the at least one assigned event, and/or (v) display, via the application, the care schedule to the at least one caregiver on at least one device associated with the at least one caregiver. The instructions may direct additional, less, or alternate functionality, including that discussed elsewhere herein.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0008] The Figures described below depict various aspects of the systems and methods disclosed therein. It should be understood that each Figure depicts an embodiment of a particular aspect of the disclosed systems and methods, and that each of the Figures is intended to accord with a

possible embodiment thereof. Further, wherever possible, the following description refers to the reference numerals included in the following Figures, in which features depicted in multiple Figures are designated with consistent reference numerals.

[0009] There are shown in the drawings arrangements which are presently discussed, it being understood, however, that the present embodiments are not limited to the precise arrangements and are instrumentalities shown, wherein:

[0010] FIG. 1 illustrates an exemplary care coordination support computer system for electronically coordinating and maintaining a care schedule of users for caregivers in accordance with the present disclosure;

[0011] FIG. 2 illustrates an exemplary configuration of an exemplary user computing device that may be used in the care coordination support computer system illustrated in FIG. 1;

[0012] FIG. 3 illustrates an exemplary configuration of an exemplary server computing device that may be used in the care coordination support computer system illustrated in FIG. 1;

[0013] FIG. 4 illustrates a flow chart of an exemplary computer-implemented method implemented by the exemplary care coordination support computer system shown in FIG. 1;

[0014] FIG. 5 illustrates a diagram of components of one or more exemplary computing devices that may be used in the care coordination support platform computer system shown in FIG. 1;

[0015] FIGS. 6-20 are screenshots of one example of a care coordination support application illustrated in FIG. 1;

[0016] FIG. 21 illustrates an exemplary computer-implemented method for coordinating care and resolving schedule conflicts via a care coordination support platform (“CCSP”) computer system; and

[0017] FIG. 22 illustrates an exemplary computer-implemented method for coordinating care and monitoring whether events have occurred as scheduled, or alternatively need to be rescheduled, via a care coordination support platform (“CCSP”) computer system.

[0018] The Figures depict preferred embodiments for purposes of illustration only. One skilled in the art will readily recognize from the following discussion that alternative embodiments of the systems and methods illustrated herein may be employed without departing from the principles of the invention described herein.

DETAILED DESCRIPTION

[0019] The present embodiments may relate to, inter alia, systems and methods for electronically coordinating a care schedule of a senior user (also referred to herein as a “user”) among one or more caregivers associated with the user. In one exemplary embodiment, the process may be performed by a care coordination support platform (“CCSP”) computer system (also referred to herein as a “CCSP platform” and a “CCSP server”).

[0020] As described below, the systems and methods described herein may leverage different types of data (e.g., user and caregiver data, user events including tasks, activities, and appointments, caregiver schedules, sensor data, and mobile device data) to determine and coordinate a care schedule of the user among one or more caregivers of the user. The systems and methods described herein may include a chatbot configured to assist in the determining and coordination of the care schedule of the user, and the chatbot may be configured to seamlessly aid caregivers in carrying out the care schedule of the user. In some embodiments, the chatbot may verbally explain scheduled events and/or schedule conflicts that may arise, and the chatbot may be configured to receive verbal responses to the chatbot from users and caregivers. In some embodiments, the user may be a senior who needs extra help going about their daily routines (e.g., rides to appointments, refills on their prescriptions, home and vehicle maintenance, emergency monitoring, and completion of tasks, like taking their medicine).

[0021] The caregivers associated with the user may include people who normally take care of the user (e.g., family members, friends, paid caregivers, etc.) and service providers of the user (e.g., health care professionals, such as doctors, nurses, physical therapists, occupational therapists, etc.).

The caregivers often have busy schedules, and it may be difficult for the caregivers to coordinate caring for the user. Accordingly, friction between caregivers may arise from constantly trying to coordinate care and scheduling to take care of the user.

[0022] The systems and methods described herein ensure that each caregiver is able to care for the user and/or carry out tasks for the user and may reduce friction between caregivers by providing a platform that automatically assigns care duties to caregivers based upon information (e.g., scheduling and calendar information) input by the caregivers and ensures that the caregivers complete their assigned duties. Further, the systems and methods described herein may learn about the user and associated caregivers and adjust interactions with the user and associated caregivers and the coordinating of the care schedule of the user based upon the learning.

EXEMPLARY USER AND CAREGIVER DATA COLLECTION

[0023] In the exemplary embodiment, a care coordination support platform (e.g., provided by a care coordination support platform server) may leverage different kinds of data (e.g., user and caregiver data, user events, caregiver schedules, sensor data, and mobile device data) to coordinate a care schedule of a user between one or more caregivers associated with the user. In the exemplary embodiment, a primary caregiver (e.g., an admin caregiver) may register for the care coordination support platform (“CCSP”) service provided by a care coordination support platform (“CCSP”) server through an application (e.g., a CCSP application) on a mobile device associated with the admin caregiver, or any other suitable device that may access the CCSP application and/or a website associated with the CCSP application.

[0024] In registering for the CCSP service, the admin caregiver may provide the CCSP server with information associated with the user. The information associated with the user may include user data (e.g., name, birthdate, height, weight, etc.), user tasks (e.g., taking medicine, bathing, eating, paying bills, getting groceries, car maintenance, home maintenance, etc.), user activities (e.g., social activities, like bingo and golfing, physical activities, like working out and keeping active, etc.), user appointments (e.g., recurring appointments like yearly physicals and bimonthly haircuts, etc.), user alert preferences (e.g., when and through which method users prefer to be alerted) and any other information associated with the user that may be useful to the CCSP server.

[0025] In other embodiments, the user may register for the service and provide the CCSP server with information themselves. Further, in registering for the CCSP service, the admin caregiver may invite other caregivers to be a part of a care team for the user. For example, the admin caregiver may provide the CCSP server with a list of emails and/or phone numbers of other caregivers who are associated with the user. The CCSP server may send an invitation link and/or code to the other caregivers instructing the other caregivers on how to sign up for the care team for the user associated with the admin caregiver.

[0026] Each caregiver, including the admin caregiver, may register themselves for the CCSP service. In registering for the CCSP service, the caregivers may provide the CCSP server with caregiver information (e.g., name, contact information, relationship to the user), caregiver schedule information (e.g., known work and/or activity schedules of the caregivers), caregiver alert preferences (e.g., when and through which method caregivers prefer to be alerted) and any other caregiver information that may be useful to the CCSP server. In some embodiments, the caregivers may link their digital calendars (e.g., provided on a mobile device associated with the caregiver) to the CCSP server such that the caregivers do not have to manually input scheduling data available to the CCSP server into the digital calendar. The user and the caregivers may update and/or edit the user and caregiver data at any time (e.g., through the CCSP application).

[0027] In the exemplary embodiment, if the caregiver is a person who normally takes care of the user and needs to view and/or be notified of the schedule of the user, the caregiver may fully register for the CCSP service. If the caregiver is a person who only provides certain services to the user (e.g., a doctor, nurse, physical therapist, occupational therapist, etc.), the caregiver may have very limited access to the CCSP service, and the CCSP server may have very limited access to the

caregiver data (e.g., the CCSP server may simply receive calendar updates from the caregiver if an event related to the user is scheduled).

[0028] In the exemplary embodiment, the CCSP server may also be configured to receive sensor data from sensors associated with the user and/or the caregivers. For example, sensors may include smart home device sensors (e.g., AMAZON ALEXA, GOOGLE HOME, and/or RING doorbells), wearable device sensors (e.g., APPLE WATCH and FITBIT), smart device sensors (e.g., smart pillboxes), sensors associated with a mobile device of the user and caregivers (e.g., GPS sensors), and any other sensors. In the exemplary embodiment, the CCSP server may be configured to store the received data (e.g., user data, caregiver data, sensor data, etc.) in a memory.

EXEMPLARY DETERMINATION OF CRITICAL CARE TASKS

[0029] In the exemplary embodiment, the CCSP server may determine whether events (including and also referred to herein as tasks, activities, and/or appointments) of the user are critical or non-critical. The CCSP server may determine that events are critical if the events are vital to the user and/or maintaining a livelihood of the user (e.g., taking medicine, helping the user get out of bed, taking the user to important appointments, helping the user bathe, etc.), and the CCSP server may determine that events are non-critical if the events are non-vital to the user and/or maintaining the livelihood of the user (e.g., mowing the user's grass, taking the user to Bingo, getting the user's oil changed, etc.).

[0030] The CCSP server may automatically determine that events of the user are critical or non-critical (e.g., through a description of the event), or the user and/or caregiver may manually input that the events of the user are critical or non-critical. Further, the determination of critical events vs. non-critical events may have an impact on user and caregiver alert and/or notification (e.g., SMS texts, emails, phone calls, and/or push button notifications from CCSP application) preferences. For example, the user and caregivers may only prefer to receive alerts and/or notifications for critical events, rather than all events.

EXEMPLARY CARE COORDINATION

[0031] In the exemplary embodiment, the CCSP server may be configured to process all of the user and caregiver data (e.g., events of the user, and schedules and preferences of the caregivers) the CCSP server receives from the user and caregivers (e.g., through the CCSP application) and coordinate a care schedule of the user between the caregivers. The CCSP server may automatically assign each task, activity, and/or appointment of the user to the caregivers based upon the received caregiver data, and the received caregiver data may include caregiver preferences and schedules. For example, one caregiver associated with the user may be a hired caregiver who is the primary caregiver of the user, and the other caregivers associated with the user may be family and/or friends of the user who may take care of events of the user if there is a need to do so (e.g., if the primary caregiver is unavailable).

[0032] The CCSP server may process the caregiver data associated with the caregivers and compare the caregiver data to the task, activity, and appointment schedule of the user. Based upon the compared data, the CCSP server may assign each task, activity, and appointment of the user to each of the caregivers. For example, the CCSP server may assign the events to the primary caregiver first (based upon a schedule of the primary caregiver) and then assign the rest of the events not assigned to the primary caregiver to other caregivers based upon the schedules of the other caregivers.

[0033] Further, the CCSP server may assign some events to the user to carry out when the user does not need assistance with the events (e.g., taking medication and/or doing daily exercises). For example, the caregivers may prefer to assign the events of the user manually, and the CCSP server may assist the caregivers in manually assigning the events (e.g., through a chatbot, described below). Once the events have been assigned to the user and/or the caregivers, the CCSP server may create a care schedule of the user. The care schedule may include all of the user's events, and the user and/or caregiver assigned to the events. The care schedule may be stored in, for example, a

care database, in a memory device associated with the CCSP server.

[0034] The CCSP server may determine if events of the care schedule of the user are taken care of by the assigned caregiver through, for example, sensor data received by the CCSP server. For instance, if a user is scheduled to take medicine at a certain time two times a day, the CCSP server may receive data from a sensor (e.g., a smart pillbox) associated with the user to determine if the pill box was opened at the certain scheduled times. Further, for example, if a caregiver is scheduled to take the user to a doctor's appointment at a certain time, the CCSP server may receive location data of the user and the caregiver (e.g., from mobile devices of the user and/or caregiver) to determine if the caregiver took the user to the doctor's appointment. Additionally, if the user is scheduled to receive a grocery delivery at a certain time, the CCSP server may receive sensor data (e.g., from a smart home device like a smart doorbell) to determine if the groceries were delivered for the user (e.g., through determining if the doorbell was rung and/or a delivery person showed up around the scheduled time).

[0035] If the CCSP server determines that a task, activity, and/or appointment has not been carried out, the CCSP server may alert (e.g., through the CCSP application) the user and/or caregivers based upon the user and caregiver data (e.g., alert preferences). Further, the CCSP server may notify the caregivers, based on the alert preferences of the caregivers, when a scheduled event has been carried out by the caregiver and/or others (e.g., service providers). In other embodiments, the user and caregivers may manually enter that the events of the user have been taken care of by the caregiver.

EXEMPLARY CARE COORDINATION SUPPORT APPLICATION

[0036] In the exemplary embodiment, a CCSP application is associated with the CCSP server. The CCSP application may be configured to receive user and caregiver data, display the care schedule of the user to the caregivers, and/or alert and/or notify the user and caregivers of assigned events. The CCSP application may be run on a device associated with the user and/or caregiver (e.g., a mobile device and/or laptop of the user or caregiver). The CCSP application may be configured to display the care schedule of the user based upon the preference of the user and caregivers. For example, the CCSP application may display a list of daily, weekly, and/or monthly tasks assigned to the user and/or caregivers, a calendar that marks when the user and/or caregiver has assigned events, and any other display method that allows the user and caregivers to easily see and interact with the care schedule of the user.

[0037] In the exemplary embodiment, the CCSP server may include a chatbot that is embedded in the CCSP application and has access to the information stored by the CCSP server (e.g., scheduled/assigned events, user data, caregiver data, etc.). The chatbot may be any suitable chatbot and/or robo-assist device that functions as described herein. The chatbot may assist the user and caregivers in adding, editing, and/or deleting user and caregiver data, coordinating care of the user between the caregivers, receiving information about the assigned care schedule, and/or receiving information about how the user and caregivers are carrying out the care schedule.

[0038] For example, instead of a user and/or caregiver having to manually input each event of the user, each notification request of the user and caregivers, and/or each schedule item of the caregivers, the user and caregivers may give instructions to the chatbot (e.g., through typing and/or speaking commands and/or questions using plain or colloquial language, rather than structured commands, into the chatbot through the CCSP application). Also, for example, if the user just added a daily medication to their routine, the user may instruct the chatbot to add the medication to the daily list of tasks for the user to carry out. Further, a caregiver may instruct the chatbot that the user's lawn needs to be mowed every week in the summer.

[0039] The chatbot may also be configured to passively assist in coordinating care for the user between the caregivers. For example, if the caregivers mostly have the care schedule of the user figured out and scheduled, the chatbot may be configured to monitor what the users and caregivers input into the chatbot and provide assistance if necessary.

[0040] For instance, if one caregiver inputs into the chatbot that the caregiver is taking the user to an appointment on Monday at 2 p.m., the chatbot may respond to the caregiver that the appointment is on their calendar. If another caregiver says that the caregiver is taking the user to breakfast on Tuesday at 10 a.m., the chatbot may respond to the caregiver that the event is not in their calendar and ask the caregiver if the caregiver would like the event added to their calendar. If the caregiver responds that the caregiver would like the event added to their calendar, the chatbot may cause the event to be added to the calendar of the caregiver.

[0041] The user and caregivers may also ask the chatbot questions, and the chatbot may, for example, convert the natural-language question of the user and caregivers into a query, run the query against a database (e.g., an event database stored in a memory device), and transmit a response to the question to the processor including an answer to the question, in response to the query returning the at least one event. For example, the user may ask the chatbot who is taking them to a haircut appointment or oil change appointment, and the caregiver may ask the chatbot to identify the last time the user had a bath.

[0042] The chatbot may also notify and/or send alerts to the user and caregivers based upon the user and caregiver alert preferences. For example, the chatbot may notify a caregiver that a user has not yet taken their medicine, and the chatbot may ask the caregiver if the caregiver would like the chatbot to send a reminder to the user to take their medicine. If the caregiver says yes, the chatbot may automatically cause the reminder to be sent to the user.

[0043] In the exemplary embodiment, the chatbot may further be configured to learn from the user and caregiver requests, responses, and/or questions. For example, if the chatbot often notifies a caregiver that the user forgets to take a nightly dose of medication, and the caregiver typically tells the chatbot to remind the user to take their medication in response to the notification from the chatbot, the chatbot may automatically cause the CCSP computing device to start reminding the user to take their nightly medicine dosage without input from the caregiver.

[0044] Further, the chatbot may be configured to verbally explain scheduled events, scheduling conflicts, and/or missed scheduled events that may arise to the user and/or caregivers. For example, if the chatbot determines that a scheduling conflict has arisen (e.g., the caregiver and/or the user are double-booked), the chatbot may verbally engage with the user and/or caregiver to explain the scheduling conflict. In verbally engaging with the user and/or caregiver, the chatbot may be configured to converse with the user and/or caregiver to resolve the scheduling conflict. Further, if the chatbot determines that a scheduled event was missed, the chatbot may verbally alert the user and/or caregiver of the missed event. In verbally alerting the user and/or caregiver, the chatbot may also be configured to converse with the user and/or caregiver to resolve and/or reschedule the missed event.

[0045] The CCSP server may further be configured to generate caregiver analytics, and the CCSP application may be configured to display the generated analytics to the user and caregivers. The CCSP server may generate activity hour, effort hour, and task distribution analytics for each caregiver and compare the analytics to the other caregivers. For example, the CCSP server may generate a chart of the time each caregiver spends caring for the user and/or the time each caregiver spends putting in effort to the care of the user for a predetermined period of time. The CCSP server may further generate a chart of a percentage of tasks for the user that each caregiver handles over the predetermined period of time.

EXEMPLARY CARE COORDINATION SUPPORT SYSTEM

[0046] FIG. 1 depicts a view of an exemplary care coordination support platform (“CCSP”) system **100** that may be used in coordinating care of a user between one or more caregivers. CCSP system **100** may include a care coordination support platform (“CCSP”) computing device **102**. In the exemplary embodiment, CCSP computing device **102** is in communication with client devices **104**, a chatbot server **106**, and a sensor server **108**. CCSP computing device **102** is also in communication with a database **118** and may communicate with database **118** through a database

server **116**.

[0047] In some embodiments, database server **116** is a component of CCSP computing device **102**. In other embodiments, database server **116** is separate from CCSP computing device **102**. Further, in some embodiments, chatbot server **106** is a component of CCSP computing device **102**. In other embodiments, chatbot server **106** is separate from CCSP computing device **102**. In some embodiments, CCSP system **100** may include a plurality of CCSP computing devices **102**, client devices **104**, chatbot servers **106**, sensor servers **108**, and/or databases **118**.

[0048] In the exemplary embodiment, CCSP computing device **102** may be configured to store user and caregiver data and generate care schedules corresponding to the user and the caregivers. CCSP computing device **102** may receive user and caregiver data from client devices **104** and use the user and caregiver data to register users and caregivers and generate care schedules for the user and caregivers. For example, a user and a caregiver may download a CCSP application **110** to a device (e.g., client device **104**) and input data into CCSP application **110** for registration with a service provided by CCSP computing device **102**. The user and caregivers may also access a website of CCSP system **100** using a web browser, and input user data into the website to register with CCSP system **100**.

[0049] The user data may include personal data (e.g., name, birthdate, height, weight, etc.), user tasks (e.g., taking medicine, bathing, eating, paying bills, getting groceries, car maintenance, home maintenance, etc.), user activities (e.g., social activities, like bingo and golfing, physical activities, like working out and keeping active, etc.), user appointments (e.g., recurring appointments like yearly physicals and bimonthly haircuts, etc.), and any other information associated with the user that may be useful to CCSP computing device **102**.

[0050] The caregiver data may include personal information (e.g., name, contact information, relationship to the user, role in caring for the user, etc.), caregiver schedule information (e.g., known work and/or activity schedules of the caregivers), caregiver preferences (e.g., which events the caregiver prefers to assist the user with and/or how often the caregiver prefers to assist the user for any period of time, like no more than three hours a week or 8 hours a month), and any other information associated with the caregivers that may be useful to CCSP computing device **102**.

[0051] CCSP application **110** may also receive other data from the user and caregivers including alert preferences of the user and caregivers (e.g., preferences of when the user and caregivers would like to be notified and how the user and caregivers would like to be notified, such as receiving a text notification and/or a push button notification from CCSP application **110**).

[0052] In the exemplary embodiment, users and caregivers may update the user and caregiver data at any time through CCSP application **110**. For example, user data that may need to be updated may include a change in and/or newly scheduled events of the user, and a change in a daily medication schedule of the user. For example, caregiver data that may need to be updated may include a change in and/or a new availability schedule of the caregiver, a new activity scheduled by the caregiver, and a vacation scheduled by the caregiver.

[0053] CCSP application **110** may be in communication with other applications of client device **104** and may import user and caregiver data from the other applications. For example, caregivers may allow CCSP application **110** to retrieve data from a calendar application of the caregivers such that the caregivers may only need to update the schedule associated with the caregiver in one application (e.g., a calendar application).

[0054] In the exemplary embodiment, CCSP computing device **102** may be configured to process all of the user and caregiver data CCSP computing device **102** receives from the user and caregivers (e.g., through CCSP application **110**) and coordinate a care schedule of the user between the caregivers. CCSP computing device **102** may automatically assign each task, activity, and/or appointment of the user to the caregivers based upon the received caregiver data. For example, four caregivers may be associated with a user. A first caregiver, who is also an admin caregiver, may be a child of the user who works a full-time job and has three children with busy schedules, a second

caregiver may be a child of the user who works part-time and does not have any children, a third caregiver may be a hired caregiver who spends up to 20 hours per week caring for the user, and a fourth caregiver may be a friend of the user who enjoys spending time with the user.

[0055] CCSP computing device **102** may process the caregiver data associated with each of the four caregivers and compare the caregiver data to the task, activity, and appointment schedule of the user. Based upon the compared data, CCSP computing device **102** may assign each task, activity, and appointment of the user to each of the caregivers and create a dynamic care schedule of the user. The admin caregiver may instruct CCSP computing device **102** (e.g., through CCSP application **110**) to assign events to the third caregiver first before assigning the events to the admin caregiver and the second and fourth caregivers. Accordingly, CCSP computing device **102** may first assign the events of the user to the third caregiver, based upon the caregiver data of the third caregiver, before assigning the tasks to the admin, second, and fourth caregivers, based upon the caregiver data of the admin, second, and fourth caregivers.

[0056] If any of the caregiver data changes (e.g., one of the caregivers has a last-minute appointment or emergency), the caregiver may notify CCSP computing device **102**, and CCSP computing device **102** may automatically assign the previously assigned task, activity, and/or appointment to another caregiver based upon the caregiver data of the other caregivers. Accordingly, CCSP computing device **102** may reduce friction in, and time spent, coordinating care of the user between caregivers by automatically assigning the care schedule of the user to the caregivers based upon caregiver data.

[0057] When CCSP computing device **102** receives user events, CCSP computing device **102** may further be configured to determine whether each of the events are critical or non-critical. For example, CCSP computing device **102** may determine whether the events are critical or non-critical based upon a description of the task, activity, and/or appointment and/or whether the user and/or caregiver specifically marks the task, activity, and/or appointment as critical or non-critical when inputting the task, activity, and/or appointment into CCSP application **110**. For example, if a description of the task, activity, and/or appointment is “doctor's appointment,” “take medicine,” and/or “help user out of bed,” CCSP computing device **102** may automatically determine that the task, activity, and/or appointment of the user is critical. However, if a description of the task, activity, and/or appointment is, for example, “Take user to bingo,” “Mow grass of user,” and/or “Get user's oil changed,” CCSP computing device **102** may automatically determine that the task, activity, and/or appointment is non-critical.

[0058] In the exemplary embodiment, CCSP computing device **102** may receive data from sensor servers **108** that are associated with sensors and may use the data to coordinate care of a user and/or determine whether assigned events of the user have been carried out. Sensor servers **108** may include smart home device servers (e.g., AMAZON and GOOGLE servers), wearable device servers (e.g., FITBIT AND APPLE servers), vehicle telematics device servers, home telematics device servers, and smart device servers. Data from the sensors may include interaction data from a smart home device (e.g., AMAZON ALEXA, GOOGLE HOME, and/or RING doorbells), activity and/or location data from a wearable device (e.g., FITBIT and APPLE WATCH), driving data and/or location data from autonomous or semi-autonomous vehicles smart vehicle sensors, and daily medication taking from a smart pillbox sensor.

[0059] Further, CCSP computing device **102** may receive sensor data (e.g., GPS location) from sensors of client devices **104**. CCSP computing device **102** and sensor servers **108** may be communicatively coupled through any suitable connection including through the Internet through many interfaces including, but not limited to, at least one of a network, such as the Internet, a local area network (LAN), a wide area network (WAN), or an integrated services digital network (ISDN), a dial-up-connection, a digital subscriber line (DSL), a cellular phone connection, and a cable modem.

[0060] CCSP computing device **102** may be configured to determine whether an assigned event of

the user has been carried out by the user and/or the caregiver. For example, if a caregiver was assigned with bringing the user dinner on a certain day, CCSP computing device **102** may use RING doorbell data of the user and/or location data of the caregiver to determine if the caregiver brought the user dinner. For another example, if a user is assigned with taking medicine three times a day at specific time intervals, CCSP computing device **102** may use smart pillbox data to determine if a pillbox of the user was opened at the specific time intervals. If CCSP computing device **102** determines that a task, activity, and/or appointment has not been carried out, CCSP computing device **102** may alert (e.g., through CCSP application **110**) the user and/or caregivers based upon the user and caregiver data (e.g., alert preferences). In some embodiments, in determining whether an event of the user has been carried out, CCSP computing device **102** may receive and/or retrieve location and time data from the event. CCSP computing device **102** may be configured to monitor and/or collect location and time data from the user (e.g., through time-stamped GPS and/or geofence location data from a mobile device, wearable device, or any other suitable device associated with the user). CCSP computing device **102** may compare the received location and time data for the event with the location and time data of the user during the event. If the received location and time data and the user location and time data are the same, CCSP computing device **102** may determine that the event was carried out. If the received location and time data and the location and time data of the user are not the same, CCSP computing device **102** may determine that the event was not carried out.

[0061] In some embodiments, CCSP computing device **102** may be configured to display (e.g., through CCSP application **110**) the generated care schedule to the user and/or caregivers. CCSP computing device **102** may display the generated care schedule to the user and caregivers through task lists, graphs, calendars, and any other suitable interface that allows the user and caregiver to easily take in and interact with the care schedule of the user.

[0062] Further, CCSP computing device **102** may be configured to generate and display (e.g., through CCSP application **110**) caregiver analytics of each caregiver based upon the care schedule of the user. For example, CCSP computing device **102** may create graphs of time each caregiver spends caring for the user, time each caregiver spends scheduling and/or participating in the scheduling of care for the user, and a task distribution of the percentage of events assigned to and carried out by each user over a predetermined period of time (e.g., a week, a month, and a year).

[0063] In the exemplary embodiment, CCSP computing device **102** may be in communication with chatbot server **106** associated with a chatbot. The chatbot may be any suitable chatbot and/or robo-assist device that functions as described herein. The chatbot may assist the user and caregivers (e.g., through CCSP application **110**) in adding, editing, and/or deleting user and caregiver data, receiving information about the assigned care schedule, and/or receiving information about how the user and caregivers are carrying out the care schedule.

[0064] For example, instead of a user and/or caregiver having to manually input (e.g., using a structured command format or menu) each task, activity, and appointment of the user, each notification request of the user and caregivers, and/or each schedule item of the caregivers, the user and caregivers may give instructions to the chatbot (e.g., through typing and/or speaking plain-language commands and/or questions into the chatbot through CCSP application **110**). Also, for example, if the user just made an appointment, the user may instruct the chatbot using natural language to add the appointment to the user's calendar and assign the appointment to one of the caregivers. Further, a caregiver may instruct the chatbot to remind the caregiver to pay the user's bills ever month.

[0065] The user and caregivers may also ask the chatbot questions. For example, the user may ask the chatbot who is taking them to Bingo on a particular day, and the caregiver may ask the chatbot to identify the last time the user took their medicine. The chatbot may also notify and/or send alerts to the user and caregivers based upon the user and caregiver alert preferences. Also, for example, the chatbot may notify a caregiver that a user has not yet taken their medicine, and the chatbot may

ask the caregiver if the caregiver would like the chatbot to send a reminder to the user to take their medicine. If the caregiver says yes, the chatbot may automatically cause the CCSP computing device **102** to send the reminder to the user.

[0066] Further, chatbot server **106**, along with CCSP computing device **102**, may be configured to use machine learning algorithms and/or programs to learn from the user and/or caregivers, as described herein. For example, CCSP computing device **102** may receive task monitoring and/or completion data generated by sensor servers **108** based upon a task being completed by the user and/or the assigned caregiver. Although described as implemented on chatbot server **106** separate from CCSP computing device **102**, the chatbot may alternatively be co-implemented on CCSP computing device **102**.

[0067] In the exemplary embodiment, client devices **104** may be computers that include a web browser or a software application, which enables client devices **104** to access remote computer devices, such as CCSP computing device **102**, using the Internet or other network. More specifically, client devices **104** may be communicatively coupled to the Internet through many interfaces including, but not limited to, at least one of a network, such as the Internet, a local area network (LAN), a wide area network (WAN), or an integrated services digital network (ISDN), a dial-up-connection, a digital subscriber line (DSL), a cellular phone connection, and a cable modem. Client devices **104** may be any device capable of accessing the Internet including, but not limited to, a desktop computer, a laptop computer, a personal digital assistant (PDA), a cellular phone, a smartphone, a tablet, a phablet, wearable electronics, smart watch, or other web-based connectable equipment or mobile devices. Further, CCSP computing device **102** may be communicatively coupled to client devices **104** and may receive information from client devices **104**.

[0068] Database server **116** may be communicatively coupled to database **118** that stores data. In one embodiment, database **118** may include user data, caregiver data, sensor data, mobile device data, assignment data, and notification data. In the exemplary embodiment, database **118** may be stored remotely from CCSP computing device **102**. In some embodiments, database **118** may be decentralized. In the exemplary embodiment, a user and/or caregiver, may access database **118** via their respective client devices **104** by logging onto CCSP computing device **102**, as described herein.

[0069] In the exemplary embodiment, client devices **104** include CCSP application **110** and a user interface **112**. User interface **112** may be used, for example, to receive notifications from CCSP computing device **102** and/or to input and verify information to be sent to CCSP computing device **102**. CCSP application **110** may be, for example, a program or application that runs on client device **104**.

EXEMPLARY USER COMPUTER DEVICE

[0070] FIG. 2 illustrates an exemplary configuration **200** of an exemplary user computing device **202**. In some embodiments, user computing device **202** may be in communication with a care coordination support platform computing device (such as CCSP computing device **102**, shown in FIG. 1). User computing device **202** may be representative of, but is not limited to client devices **104** and/or sensor servers **108**. For example, user computing device **202** may be a mobile device, smartphone, tablet, smartwatch, wearable electronic, laptop, desktop, or another type of computing device associated with an account holder (e.g., the user and/or the associated caregivers).

[0071] User computer device **202** may be operated by a user **204** (e.g., a user of CCSP system **100**, shown in FIG. 1 and substantially similar to the user and/or the caregivers described herein). User computer device **202** may receive input from user **204** via an input device **214**. User computer device **202** includes a processor **208** for executing instructions. In some embodiments, executable instructions may be stored in a memory area **210**. Processor **208** may include one or more processing units (e.g., in a multi-core configuration). Memory area **210** may be any device allowing information such as executable instructions and/or user and registration data to be stored and

retrieved. Memory area **210** may include one or more computer-readable media.

[0072] User computer device **202** also may include at least one media output component **212** for presenting information to user **204**. Media output component **212** may be any component capable of conveying information to user **204** and may be used to at least partially implement user interface **112** (shown in FIG. **1**). In some embodiments, media output component **212** may include an output adapter (not shown), such as a video adapter and/or an audio adapter. An output adapter may be operatively coupled to processor **208** and operatively coupleable to an output device, such as a display device (e.g., a cathode ray tube (CRT), liquid crystal display (LCD), light emitting diode (LED) display, or “electronic ink” display) or an audio output device (e.g., a speaker or headphones).

[0073] In some embodiments, media output component **212** may be configured to present a graphical user interface (e.g., a web browser and/or a client application) to user **204**. A graphical user interface may include, for example, care calendars for the user and/or associated caregivers, notifications for the user and/or associated caregivers, assigned tasks of the caregivers, an activity analytics of the caregivers, and/or a messaging page for interacting with a chatbot (e.g., the chatbot associated with chatbot server **106** shown in FIG. **1**).

[0074] In some embodiments, user computer device **202** may include input device **214** for receiving input from user **204**. User **204** may use input device **214** to, without limitation, interact with CCSP system **100** (e.g., using CCSP application **110**), CCSP computing device **102**, or any of client devices **104**, chatbot servers **106**, and sensor servers **108** (shown in FIG. **1**). Input device **214** may include, for example, a keyboard, a pointing device, a mouse, a stylus, and/or a touch sensitive panel (e.g., a touch pad or a touch screen) and may be used to at least partially implement user interface **112** (shown in FIG. **1**). A single component, such as a touch screen, may function as both an output device of media output component **212** and input device **214**. User computer device **202** may further include at least one sensor, including, for example, a gyroscope, an accelerometer, a position detector, a biometric input device, and/or an audio input device. In some embodiments, at least some data collected by user computer device **202** may be transmitted to CCSP computing device **102**. In the exemplary embodiment, data collected by user computer device **202** may be included in user and caregiver data.

[0075] User computer device **202** may also include a communication interface **216**, communicatively coupled to any of CCSP computing device **102**, client devices **104**, chatbot servers **106**, and sensor servers **108**. Communication interface **216** may include, for example, a wired or wireless network adapter and/or a wireless data transceiver for use with a mobile telecommunications network.

[0076] Stored in memory area **210** may be, for example, computer-readable instructions for providing a user interface to user **204** via media output component **212** and, optionally, receiving and processing input from input device **214**. The user interface may include, among other possibilities, a web browser and/or a client application. Web browsers enable users, such as user **204**, to display and interact with media and other information typically embedded on a web page, a website, or an application hosted by CCSP computing device **102** and/or client device **104**. A client application may allow user **204** to interact with, for example, any of CCSP computing device **102**, client devices **104**, chatbot servers **106**, and sensor servers **108**. For example, instructions may be stored by a cloud service and the output of the execution of the instructions sent to the media output component **212**. User computing device **200** may include additional, less, or alternate functionality, including that discussed elsewhere herein.

EXEMPLARY SERVER DEVICE

[0077] FIG. **3** depicts an exemplary configuration **300** of an exemplary server computer device **302**, in accordance with one embodiment of the present disclosure. Server computer device **302** may include, but is not limited to, CCSP computing device **102** (shown in FIG. **1**) and/or chatbot server **106** (shown in FIG. **1**). Server computer device **302** may include a processor **305** for

executing instructions. Instructions may be stored in a memory area **310**. Processor **305** may include one or more processing units (e.g., in a multi-core configuration).

[0078] Processor **305** may be operatively coupled to a communication interface **315** such that server computer device **302** may be capable of communicating with a remote device such as another server computer device **302** or a user computing device, such as client device **104** (shown in FIG. 1). For example, communication interface **315** may receive requests from or transmit requests to client devices **104** via the Internet.

[0079] Processor **305** may also be operatively coupled to a storage device **325**. Storage device **325** may be any computer-operated hardware suitable for storing and/or retrieving data, such as, but not limited to, data associated with database **118** (shown in FIG. 1). In some embodiments, storage device **325** may be integrated in server computer device **302**. For example, server computer device **302** may include one or more hard disk drives as storage device **325**. In other embodiments, storage device **325** may be external to server computer device **302** and may be accessed by a plurality of server computer devices **302**. For example, storage device **325** may include a storage area network (SAN), a network attached storage (NAS) system, and/or multiple storage units such as hard disks and/or solid state disks in a redundant array of inexpensive disks (RAID) configuration.

[0080] In some embodiments, processor **305** may be operatively coupled to storage device **325** via a storage interface **320**. Storage interface **320** may be any component capable of providing processor **305** with access to storage device **325**. Storage interface **320** may include, for example, an Advanced Technology Attachment (ATA) adapter, a Serial ATA (SATA) adapter, a Small Computer System Interface (SCSI) adapter, a RAID controller, a SAN adapter, a network adapter, and/or any component providing processor **305** with access to storage device **320**.

[0081] Processor **305** executes computer-executable instructions for implementing aspects of the disclosure. In some embodiments, processor **305** may be transformed into a special purpose microprocessor by executing computer-executable instructions or by otherwise being programmed.

EXEMPLARY COMPUTER-IMPLEMENTED METHOD

[0082] FIG. 4 depicts a flow chart illustrating a computer-implemented method **400** for coordinating care between caregivers associated with a user. In the exemplary embodiment, method **400** may be implemented by a care coordination support platform computer system such as CCSP computing device **102** (shown in FIG. 1).

[0083] Method **400** may include registering **402** a user and registering **404** at least one caregiver associated with the user for a care coordination support platform service (e.g., provided by CCSP computer system shown in FIG. 1). Method **400** may also include receiving **406** at least one event associated with the user.

[0084] Method **400** may further include assigning **408** the at least one event to the at least one caregiver and creating **410** a care schedule of the user that includes the assigned **408** at least one event. Method **400** may further include displaying **410**, via an application associated with CCSP computing device **102**, shown in FIG. 1, the care schedule to the at least one caregiver.

EXEMPLARY COMPUTER DEVICE

[0085] FIG. 5 depicts a diagram **500** of components of one or more exemplary computing devices **510** that may be used in care coordination support platform system **100** (shown in FIG. 1). In some embodiments, computing device **510** may be similar to CCSP computing device **102** (shown in FIG. 1). Database **520** may be coupled with several separate components within computing device **510**, which perform specific tasks. In this embodiment, database **520** may include user data **521**, caregiver data **522**, sensor data **523**, calendar data **524**, task data **525**, assignment data **526**, and alert data **527**. In some embodiments, database **520** is similar to database **118** (shown in FIG. 1).

[0086] Computing device **510** may include database **520**, as well as data storage devices **530**. Computing device **510** may also include an analytics component **540** for analyzing received user and caregiver data to assign events to caregivers based upon the received data. Computing device **510** may further include application/display component **550** for generating and displaying

information to users, such as through CCSP application **110** (shown in FIG. 1), and supporting CCSP application **110**. Moreover, computing device **510** may include communications component **560** for receiving and transmitting data (e.g., to and from user devices **104** and chatbot server **106**), such as user data **521**, caregiver data **522**, sensor data **523**, calendar data **524**, task data **525**, assignment data **526**, and alert data **527**. Computing devices **510** may include additional, less, or alternate functionality, including that discussed elsewhere herein.

EXEMPLARY CARE COORDINATION SUPPORT APPLICATION

[0087] FIGS. **6-20** include screenshots of one example embodiment of an application (e.g., CCSP application, shown in FIG. 1) executable as part of a care coordination support platform system (e.g., care coordination support platform system **100**, shown in FIG. 1). The application may be accessible on any suitable electronic device, such as a mobile phone, tablet, watch, or any other computing device. The application enables a user to determine what tasks the users need to complete, view a care coordination schedule of the user, enables one or more caregivers to add tasks for the user, view the tasks that the one or more caregivers have to complete for and/or with the user, and coordinate the care schedule between the one or more caregivers.

[0088] In some embodiments, the application may enable the user and the one or more caregivers to subscribe to alerts, notifications, and/or reminders. Further, in some embodiments, the application may implement a chatbot that allows the user and/or the one or more caregivers to seamlessly add tasks, set reminders, and issue reminders based upon natural language instructions.

[0089] The application may be configured to communicate with various other software and/or applications on the computing devices of the users and/or the one or more caregivers. For example, the application may be able to access or otherwise communicate with calendar applications and/or contact applications. The application may be configured to retrieve data from and/or report data to these other applications. In addition, the application may be configured to track, monitor, and/or record application utilization metrics for the user and/or the caregivers, such as how often the user and/or the caregivers access the application, and the various features of the application used by the user and/or the caregivers.

[0090] In one embodiment, the application, once downloaded onto the computing device of the user and/or the caregivers, may not require internet connectivity to perform some or all of the functionality of the application (e.g., setting alerts and notifying the user and/or caregivers of the alerts). In some embodiments, all or a portion of the data input by the user and/or caregivers into the application (including, for example, application utilization metrics, task logs, etc.) may be electronically transmitted to a server (e.g., CCSP server **102**) for processing, and the processed data may be transmitted back for further processing and/or display by the application.

[0091] FIG. **6** illustrates an initial welcome page **600** that may include a header **602**. Although not specifically shown, header **602** may include a home button, a back button, and any other buttons to help the user and/or the caregivers navigate the application. In some embodiments, initial welcome page **600** may further include a footer (not specifically shown) that may include additional buttons to help the user and/or the caregiver navigate the application. Initial welcome screen **600** may also include a “Register a New Care Team” button **604** that, when clicked, may cause the application to display a user registration screen **700**.

[0092] FIG. **7** illustrates user registration page **700** for the application. User registration page **700** may include a first field **702** for the user and/or an administrative caregiver (e.g., “admin caregiver”) to enter the name of the user and a second field **704** to, if user registration page **700** is filled out by the admin caregiver, enter the relationship of the admin caregiver to the user.

[0093] Further, user registration page **700** may include certain user needs **706**, and each user need **706** may be accompanied by a corresponding check box **708** that the user and/or admin caregiver may check if the user has the corresponding user need **706**. For example, if the user has a “bathing” need **706**, the user and/or admin caregiver may check check box **708** that corresponds to the “bathing” need **706**.

[0094] User registration page **700** may further include a third field **710** for the user and/or admin caregiver to fill in additional needs that the user may have that are not spelled out in user needs **706**. Further, user registration page **700** may include a fourth field **712** for the user and/or admin caregiver to enter email addresses, phone numbers, and/or other contact information of other caregivers such that the application may invite the other caregivers to download the application.

[0095] User registration page may include an “Invite” button **714**, that, when clicked, may cause the application to invite the other caregivers to download the application and/or show a caregiver registration page **800**, a “Save” button **716** that, when clicked, may cause the application to save the information in fields **702**, **704**, **710**, and **712**, and a “Cancel” button **718** that, when clicked may cause the application to go back to initial welcome page **600**. Once other caregivers download the application, the other caregivers may be directly navigated to caregiver registration page **800** and may bypass initial welcome page **600** and/or user registration page **700**.

[0096] FIG. **8** illustrates caregiver registration page **800** for the application that allows the admin caregiver and/or the other caregivers to register for the application. Caregiver registration page **800** may include a first field **802** for the caregiver to enter the name of the caregiver, a second field **804** for the caregiver to enter a password for the application, a third field **806** for the caregiver to enter a role that the caregiver carries out in caring for the user, a fourth field **808** for the caregiver to enter a relationship of the caregiver to the user, and a fifth field **810** for the caregiver to enter contact information (e.g., phone number(s), email address, and/or home address).

[0097] Further, caregiver registration page **800** may include a sixth field **812** for the caregiver to enter availability (e.g., work schedule and/or calendar schedule) of the caregiver. The availability of the caregiver may be manually entered into the sixth field **812** and/or the caregiver may click an “Import Calendar” button **814** that, when clicked, may cause the application to access calendar data of the user and automatically populate sixth field **812**. Caregiver registration page **800** may further include a “Register” button **816** that, when clicked, may cause the application to save and store the information that caregiver entered into fields **802**, **804**, **806**, **808**, **810**, and/or **812**, and display a welcome screen **900**, and caregiver registration page **800** may include a “Cancel” button **818** that, when clicked, may cause the application to close and show a different screen (e.g., a home screen) of the device running the application.

[0098] FIG. **9** illustrates welcome page **900** that may be the first page that the user and/or the admin caregiver are directed to until a user care team is complete (e.g., the user and all caregivers associated with the user are registered through the application). Welcome page **900** may include a “View User Care Team” button **902** that, when pressed, may cause the application to display a user care team page **1000** and a “Register New Care Team” button **904**, when pressed, may cause the application to display user registration page **700**.

[0099] FIG. **10** illustrates user care team page **1000** that may allow the user and/or the admin caregiver to view and/or edit the care team associated with the user. User care team page **1000** may include a list **1002** of registered members of the care team including the names of the caregivers (e.g., entered by the caregivers in first field **802** of caregiver registration page **800**, shown in FIG. **8**) and roles of the caregivers (e.g., entered by the caregivers in third field **806** of caregiver registration page **800**, shown in FIG. **8**).

[0100] User care team page **1000** may also include an “Edit” button **1006** that, when clicked, allows the user and/or admin caregiver to edit list **1002** and a “Done” button **1004** that, when clicked, causes the application to save the care team and store the care team as fully registered. User care team page **1000** may also include a first field **1008** where the user and/or the admin caregiver may manually enter contact information of additional caregivers to add to the care team.

[0101] Additionally or alternatively, the user and/or the admin caregiver may press a “Open Contacts” button **1010** that, when pressed, causes the application to open the contacts of the user and/or admin caregiver and allow the user and/or admin caregiver to automatically choose which contacts the user and/or admin caregiver would like to invite to the care team. When the user and/or

admin caregiver is done adding contact information of additional caregivers, the user and/or admin caregiver may press an “Invite” button **1012** that, when clicked, causes the application to invite the additional caregivers to register for the application.

[0102] FIG. **11** shows an example caregiver planner page **1100** that the user and/or caregivers may be directed to when the user and/or caregivers open the application after the user and/or caregivers have been registered for the application. Caregiver planner page **1100** may include a care calendar **1102** of the user. In the example embodiment, care calendar **1102** is shown in a monthly view.

[0103] Additionally or alternatively, the user and/or caregivers may change care calendar **1102** view by clicking on a month button **1104** that, when clicked, shows care calendar **1102** in a monthly view, a week button **1106** that, when clicked, shows care calendar **1102** in a weekly view, and a day button **1108** that, when clicked, shows care calendar **1102** in a daily view. Care calendar **1102** may include lines **1110** that indicate to the user and/or caregivers when a task, activity, and/or appointment is scheduled for the user. For example, lines **1110** may indicate that the caregiver using the application has assigned events to take care of for and/or with the user on specific dates of care calendar **1102**, or lines **1110** may indicate that the user has assigned events on specific dates of care calendar **1102**, and lines **1110** may not be specific to one caregiver.

[0104] Further, caregiver planner page **1100** may include a “User's Care Activities” button **1112** that, when clicked, may cause the application to display user's care activities page **1900**. Caregiver planner page **1100** may also include a chatbot button **1114** that, when clicked, causes the application to open a chatbot messaging page, examples of which are shown in a first, a second, a third, and a fourth chatbot messaging page **1400**, **1500**, **1600**, **1700**, and **1800**.

[0105] FIG. **12** illustrates another example of a caregiver planner page **1200**. Caregiver planner page **1200** is substantially similar to caregiver planner page **1100** (shown in FIG. **11**). However, caregiver planner page **1200** shows to the caregiver using the application their daily assigned tasks **1202** above care calendar **1102**. Daily assigned tasks **1202** may be accompanied by check boxes **1204**. The caregiver may manually check check boxes **1204**, or cause them to be checked by instructing the chatbot, when the accompanying daily task **1202** is completed by the caregiver, and/or the application may automatically check check boxes **1204** after the application determines that the accompanying daily task **1202** is completed by the caregiver and/or user (e.g., via sensor data and/or mobile device data, as described above).

[0106] FIG. **13** illustrates first chatbot messaging page **1300**. First chatbot messaging page **1300** may include a first icon **1302** for caregiver messages **1306** and a second icon **1304** for chatbot messages **1308**. In the exemplary embodiment, chatbot messaging page **1300** includes a first message field **1310** where the user and/or caregivers may enter messages for a chatbot (e.g., chatbot **106**, shown in FIG. **1**) and a send icon **1312** that, when clicked, may cause the application to send the entered message to the chatbot.

[0107] In other embodiments, messages may be entered into the chatbot through the user and/or caregiver speaking into a microphone of the device associated with the user and/or caregiver, or through any other suitable media such that the chatbot can receive and interpret the messages. Chatbot messaging page **1300** shows the caregiver asking the chatbot to add an appointment to the user's calendar (e.g., care calendar **1102** shown in FIG. **1**). Further, chatbot messaging page **1300** shows a conversation between a caregiver and the chatbot. In other embodiments, multiple caregivers and/or the user may be involved in the conversation with the chatbot.

[0108] FIGS. **14**, **15**, **16**, **17**, and **18** illustrate second, third, fourth, fifth, and sixth chatbot messaging pages **1400**, **1500**, **1600**, **1700**, and **1800**. Chatbot messaging pages **1400**, **1500**, **1600**, **1700**, and **1800** are substantially similar to first chatbot messaging page **1300** (shown in FIG. **13**) and are illustrated to show more capabilities of the chatbot. For example, chatbot messaging page **1400** shows the chatbot setting reminders for the caregiver to pay the user's bills, chatbot messaging page **1500** shows the chatbot answering the caregiver's question about when the user takes medicine and asking the caregiver if the chatbot needs to send a reminder to the user to take

the medicine of the user, chatbot messaging page **1600** shows the chatbot notifying the caregiver that the user has not yet taken the medicine of the user at the scheduled time, chatbot messaging page **1700** shows the chatbot assisting two caregivers in coordinating and scheduling who is taking the user to doctor's appointments, and chatbot messaging page **1800** shows the chatbot assisting caregivers in determining who can take the user to dinner since the scheduled caregiver has a conflict.

[0109] Chatbot messaging pages **1400**, **1500**, **1600**, **1700**, and **1800** further show that the chatbot may learn and predict what the user and/or caregiver might ask the chatbot to do. For example, chatbot messaging page **1400** shows the chatbot asking the user if the reminder to pay the user's bills is a repeating task. Since bills are a task that must be completed frequently and since the caregiver asked the chatbot to set a reminder for the caregiver to pay bills, the chatbot may predict that the caregiver wants a frequent reminder to pay the bills. Further, chatbot messaging page **1800** shows that the first caregiver informs the chatbot and other caregivers that the first caregiver cannot take the user to dinner, and the second caregiver asks what time. The chatbot may predict that since the second caregiver asked the time, that the second caregiver is asking if the second caregiver would be able to take the user to dinner and may learn to check the second caregiver's schedule to determine if the second caregiver is busy. Accordingly, the chatbot may learn what the caregivers are asking and/or needing and answer the questions and/or fulfill the caregivers' needs without needing input from the caregivers.

[0110] FIG. **19** illustrates user's care activities page **1900**. User's care activities page **1900** may include a list **1908** of the user's scheduled activities along with the caretaker assigned to carry out the scheduled activity for and/or with the user and a time at which the user's activity is scheduled. List **1908** may be accompanied by check boxes **1910**. The assigned caregiver, the user, and/or the admin caregiver may manually check, or may instruct the chatbot to check, the corresponding check box **1908** when the activity has been taken care of by the assigned caregiver and/or the application may automatically check the corresponding check box when the application determines that the activity has been taken care of, as described above.

[0111] List **1908** may be viewed by the user and/or caregivers on a daily basis (i.e., as a daily schedule) by clicking a day button **1902**, on a weekly basis by clicking a week button **1904**, and/or on a monthly basis by clicking a month button **1906**. User's care activities page **1900** may further include an "Add Activity" button **1912** that, when clicked, allows the user and/or caregiver to manually add an activity or add an activity through the chat bot (e.g., via a chatbot messaging page **1400**, **1500**, **1600**, **1700**, and **1800**, shown in FIGS. **13**, **14**, **15**, **16**, **17**, and **18**) and a "View Trends" button **1914** that, when clicked, may cause the application to display an activity trends page **2000**.

[0112] FIG. **20** illustrates activity trends page **2000**. Activity trends page **2000** may include an activity hours chart **2002**, an effort hours chart **2004**, and/or a distribution chart **2006**. Activity hours chart **2002** may display the amount of time that each caregiver spends carrying out activities for the user. Effort hours chart **2004** may display the amount of time that each caregiver spends putting in effort to the care of the user (e.g., checking and/or coordinating care, general time spent with the user when the caregiver is not scheduled, etc.). Distribution chart **2006** may display the percentage of tasks (e.g., determined by time spent doing tasks and/or the total number of tasks assigned to and carried out by the caregiver) that each caregiver handles for the user.

[0113] Charts **2002**, **2004**, and **2006** may assist the user and/or caregivers in determining if the assigning of tasks is fair and if one or more caregivers are not contributing sufficiently to the care team. Activity trends page **2000** may further include a "Back" button **2008** that, when clicked, causes the application to display user's care activities page **1900**.

[0114] In other embodiments, the application may include additional features and functionality. For example, the application may present a user interface to the user and/or caregivers including an option for the user and/or caregivers to view or input additional data to their profile. The application may additionally provide an option for the user and/or caregivers to input, view, and/or

edit medication information for the user. For example, the user and/or caregivers may be able to see the user's daily medication schedule and determine if the user is taking the medication (e.g., through a sensor, as described above).

EXEMPLARY CARE COORDINATION SUPPORT PLATFORM

[0115] In one aspect, a care coordination support platform computer system including at least one processor in communication with at least one memory device may coordinate care. The processor may be programmed to: (i) register a user through an application, wherein the user inputs personal and scheduling data into the application, (ii) register a caregiver associated with the user through the application, wherein the caregiver inputs personal and scheduling data into the application, and wherein the caregiver is one caregiver out of a plurality of caregivers associated with the user, (iii) receive input from at least one of the user and the caregiver defining at least one event associated with the user, wherein the event includes at least one of a task, an activity, and an appointment, (iv) assign the at least one event to at least one caregiver of the plurality of caregivers based upon the personal and scheduling data of the plurality of caregivers, (v) create a care schedule of the user, wherein the care schedule of the user includes the at least one assigned event, and/or (vi) display, via the application, the care schedule to the at least one caregiver on at least one device associated with the at least one caregiver.

[0116] The care coordination support platform may further include a chatbot in communication with the at least one processor, and the processor may be programmed to: (i) receive, via the chatbot, a question regarding the at least one event from at least one of the user and the caregiver, wherein the question is at least one of a natural-language typed question and a natural-language voice question, (ii) convert, via the chatbot, the question into a query, (iii) run, via the chatbot, the query against an event database stored in the at least one memory device, (iv) transmit, via the chatbot, a response to the question to the processor, wherein the response includes an answer to the question, in response to the query returning the at least one event, (v) receive, via the chatbot, input from at least one of the user and the caregiver defining a second event associated with the user, wherein the input is at least one of a natural-language typed questions and a natural-language voice question, and/or (vi) add the second event to the care schedule of the user.

[0117] The processor may further be programmed to: (i) receive sensor data from sensors associated with at least one of the user and the caregiver, wherein the sensors include at least one of mobile device sensors, vehicle telematics sensors, home telematics sensors, and smart home device sensors, (ii) determine, based upon the received sensor data, whether the at least one assigned event has been carried out by the respective caregiver of the plurality of caregivers, and/or (iii) notify the at least one caregiver of the plurality of caregivers whether the at least one assigned event has been carried out by the respective caregiver of the plurality of caregivers.

[0118] The processor may further be programmed to: (i) update, via the application, a calendar of the respective caregiver of the plurality of caregivers in response to the at least one event being assigned to the caregiver and/or (ii) display, via the application and based upon the calendar, a daily schedule including the at least one event assigned to the respective caregiver for each particular day.

[0119] The processor may further be programmed to: (i) determine whether the at least one event is critical to the user, (ii) receive alert preferences from the user and the caregiver, wherein the alert preferences include at least one of a preferred method of receiving an alert, whether the user and the caregiver prefer to receive alerts for all of the at least one event, whether the user and the caregiver prefer to receive alerts only for critical ones of the at least one event, and how often the user and the caregiver prefer to be alerted, (iii) transmit alerts to the user and the caregiver based upon the respective alert preferences of the user and the caregiver, (iv) monitor an active time during which each caregiver of the plurality of caregivers cares for the user, (v) store the monitored active time for each caregiver, (vi) compare the active time for each caregiver with the active time for each other caregiver, (vii) display the compared active time to each caregiver of the plurality of

caregivers, wherein the compared active time is displayed for a time interval, and wherein the time interval is at least one of a week, two weeks, three weeks, a month, and a year, (viii) provide rewards for each caregiver of the plurality of caregivers based upon the active time that each caregiver cares for the user, wherein the rewards include at least one of virtual badges, virtual tokens, virtual stickers, and monetary awards, (ix) receive medication data associated with the user from one or more of the user and the caregiver, (x) provide, via the application, a daily medication list, and/or (xi) allow, via the application, the one or more of the user and the caregiver to indicate when the user takes each medication from the daily medication list.

[0120] The personal and scheduling data of the user may include user data including a name, a birthdate, a height, and a weight, user tasks including taking medicine, bathing, eating, paying bills, getting groceries, car maintenance, and home maintenance, user activities including social activities and physical activities, and user appointments including recurring appointments. The personal and scheduling data of the caregiver may include caregiver data including a name, a relationship to the user, a role in the caring of the user, and caregiver schedule information including known work and activity schedules.

[0121] The foregoing CCSP computer systems may include additional, less, or alternate functionality, including that discussed elsewhere herein. For instance, to coordinate care, the at least one processor may be configured to: input user and caregiver data into a machine learning algorithm, model, or program trained to coordinate care based upon user and caregiver data.

EXEMPLARY SCHEDULE CONFLICT RESOLUTION

[0122] FIG. **21** illustrates an exemplary computer-implemented method for coordinating care and resolving schedule conflicts via a care coordination support platform (“CCSP”) computer system **2100**. The CCSP computer system may include at least one processor and/or associated transceiver in communication with at least one memory device.

[0123] The computer-implemented method **2100** may include, via one or more processors and/or associated transceivers, building, creating, and/or otherwise generating **2102** an electronic calendar for a senior with multiple tasks, appointments, and/or activities (or “events,” as used herein), with multiple caregivers (etc., professional caregivers, service providers, medical offices, family members, friends, etc.) having electronic access to the calendar. The generating **2102** of the senior's calendar may include the registering user and caregiver functionality; defining event (such as tasks, appointments, and/or activities, as discussed herein) functionality; and/or assigning event functionality as discussed elsewhere herein.

[0124] The computer-implemented method **2100** may include, via one or more processors and/or associated transceivers, collecting **2104** updated schedule information, mobile device data, sensor data (such as wearable sensor data, smart home sensor data, and/or smart vehicle sensor data), social media data, and/or other schedule-related data associated with the multiple caregivers having access to the electronic calendar and/or the senior.

[0125] The computer-implemented method **2100** may include, via one or more processors and/or associated transceivers, analyzing **2106** the updated schedule-related data to determine whether any schedule conflicts exist. For instance, the method **2100** may include, via one or more processors and/or associated transceivers, inputting the updated schedule-related data into a machine learning algorithm, model, module, or program trained to identify schedule or calendar conflicts based upon calendar and/or schedule-related data associated with a senior and multiple caregivers and/or service providers.

[0126] When a schedule conflict is identified, the computer-implemented method **2100** may include, via one or more processors and/or associated transceivers, analyzing **2108** the schedule-related data to determine a resolution to the schedule conflict. For instance, the method **2100** may include, via one or more processors and/or associated transceivers, inputting the updated schedule-related data into a machine learning program, model, module, or algorithm trained to determine calendar conflict resolutions using updated schedule-related data associated with the senior and/or

multiple caregivers or service providers.

[0127] The computer-implemented method **2100** may include, via one or more processors and/or associated transceivers, invoking **2110** a calendar chatbot to orally or verbally explain the schedule conflict to the senior and/or caregivers, and orally or verbally propose the scheduling conflict resolution identified.

[0128] The computer-implemented method **2100** may include, via one or more processors and/or associated transceivers, receiving **2112** acceptance or acknowledgement of the scheduling change/conflict resolution from the senior and/or caregivers, such as via wireless communication or data transmission with the mobile devices, or other computing devices, of the senior and/or caregivers.

[0129] Additionally or alternatively, when a schedule conflict is identified, the computer-implemented method **2100** may include, via one or more processors and/or associated transceivers, invoking **2114** the calendar chatbot to orally or verbally engage the senior and/or one or more caregivers, such as via their mobile or other computing devices, to explain the schedule conflict.

[0130] The computer-implemented method **2100** may then include, via one or more processors and/or associated transceivers, engaging **2116** in a conversation with the senior and/or one or more caregivers via the calendar chatbot to arrive at a calendar conflict resolution. For instance, the one or more processors may analyze the updated schedule-related data to identify potential replacement caregivers for a scheduled appointment or activity, or to identify potential replacement times for a given appointment or activity with a specific caregiver. Additionally or alternatively, the calendar chatbot may ask the senior and/or a caregiver for one or more replacement times and/or alternate caregivers for an event (e.g., task, activity, or appointment) that has to be rescheduled.

[0131] After which, the computer-implemented method **2100** may include, via one or more processors and/or associated transceivers, updating **2118** the senior's calendar, and transmitting electronic notifications of the scheduling/calendar change to the mobile or other computing devices of the senior and/or caregivers. The computer-implemented method **2100** may include additional, less, or alternate actions, including those discussed elsewhere herein.

[0132] In one aspect, a care coordination support platform (“CCSP”) computer system for coordinating care may be provided. The CCSP computer system may include at least one processor and/or associated transceiver in communication with at least one memory device. The at least one processor and/or associated transceiver may be programmed to: (1) register a senior (or other user) through an application, wherein the senior inputs personal and scheduling data into the application; (2) register multiple caregivers associated with the senior through the application, wherein each caregiver inputs personal and scheduling data into the application; (3) receive input from at least one of the senior and the caregiver defining at least one event associated with the senior, wherein the event includes at least one of a task, an activity, and an appointment; (4) assign the at least one event to at least one caregiver of the multiple caregivers based upon the personal and scheduling data of the multiple caregivers; (5) create a care schedule of the senior, wherein the care schedule of the senior includes the at least one assigned event; (6) receive updated scheduling-related data associated with the senior and/or the multiple caregivers; and/or (7) compare the care schedule of the senior with the updated scheduling-related data to determine a calendar conflict associated with the senior and/or at least one caregiver. The CCSP computer system may include additional, less, or alternate functionality, including that discussed elsewhere herein.

[0133] For instance, the at least one processor and/or associated transceiver may be further configured to invoke a chatbot to verbally explain to the senior or a caregiver that a calendar conflict exists. The at least one processor and/or associated transceiver may be further configured to analyze the care schedule of the senior with the updated scheduling-related data to determine an alternate caregiver that is available at the time and date of the calendar conflict to resolve the calendar conflict. The at least one processor and/or associated transceiver may be further configured to invoke a chatbot to verbally explain to the senior or a caregiver that a calendar

conflict exists, and verbally explain a resolution to the calendar conflict involving the alternate caregiver.

[0134] The at least one processor and/or associated transceiver may be further configured to determine a resolution to the calendar conflict from processor analysis of the schedule-related data, and further configured to invoke a chatbot to verbally explain to the senior or a caregiver that a calendar conflict exists, and verbally explain the resolution to the calendar conflict determined from analysis of the schedule-related data.

[0135] The at least one processor and/or associated transceiver may be further configured to input the schedule-related data into a machine learning algorithm, model, module, or program that is trained to identify calendar or scheduling conflicts and/or associated scheduling resolutions using schedule-related and/or calendar data associated with seniors and multiple caregivers.

[0136] The at least one event may be a medical-related event or a doctor's appointment; taking a prescribed medication at a given time; an exercise or physical fitness event; a home delivery; a grocery delivery; a vehicle or home repair event; and/or other events, including those discussed elsewhere herein.

[0137] In another aspect, a computer-implemented method for coordinating care and resolving schedule conflicts may be provided. The computer-implemented method may be performed by a care coordination support platform ("CCSP") computer system including at least one processor and/or associated transceiver in communication with at least one memory device. The computer-implemented method may include (1) registering a senior (or other user) through an application, wherein the senior inputs personal and scheduling data into the application; (2) registering multiple caregivers associated with the senior through the application, wherein each caregiver inputs personal and scheduling data into the application; (3) receiving input from at least one of the senior and the caregiver defining at least one event associated with the senior, wherein the event includes at least one of a task, an activity, and an appointment; (4) assigning the at least one event to at least one caregiver of the multiple caregivers based upon the personal and scheduling data of the multiple caregivers; (5) creating a care schedule of the senior, wherein the care schedule of the senior includes the at least one assigned event; (6) receiving updated scheduling-related data associated with the senior and/or the multiple caregivers; and/or (7) comparing the care schedule of the senior with the updated scheduling-related data to determine a calendar conflict associated with the senior and/or at least one caregiver. The method may include additional, less, or alternate actions, including those discussed elsewhere herein.

[0138] For instance, the method may include invoking a chatbot to verbally explain to the senior or a caregiver that a calendar conflict exists. The method may further include analyzing the care schedule of the senior with the updated scheduling-related data to determine an alternate caregiver that is available at the time and date of the calendar conflict to resolve the calendar conflict. The method may further include invoking a chatbot to verbally explain to the senior or a caregiver that a calendar conflict exists, and/or verbally explain a resolution to the calendar conflict involving the alternate caregiver.

[0139] The method may further include determining a resolution to the calendar conflict from processor analysis of the schedule-related data, and invoking a chatbot to verbally explain to the senior or a caregiver that a calendar conflict exists, and verbally explain the resolution to the calendar conflict determined from analysis of the schedule-related data.

[0140] The method may further include inputting the schedule-related data into a machine learning algorithm, model, module, or program that is trained to identify calendar or scheduling conflicts and/or associated scheduling resolutions using schedule-related and/or calendar data associated with seniors and multiple caregivers.

[0141] In another aspect, a care coordination support platform ("CCSP") computer system for coordinating care and resolving schedule conflicts may be provided. The CCSP computer system may include at least one processor and/or associated transceiver in communication with at least one

memory device. The at least one processor and/or associated transceiver may be programmed to: (1) build, create, or otherwise generate a senior calendar (for a senior or other user) (i) with tasks, appointments, and/or activities (and/or other events) and (ii) with multiple caregivers having access to the senior calendar, wherein individual caregivers are assigned responsibilities for certain tasks, appointments, and/or activities; (2) collect updated schedule information, mobile device data, sensor data (such as home, vehicle, or wearable sensor data), social media data, and/or other updated schedule-related data associated with the multiple caregivers and/or the senior (such as via wireless communication and/or data transmission over one or more radio frequency links); and/or (3) analyze the updated schedule-related data to determine a schedule conflict to facilitate coordinating care for the senior. The CCSP computer system may include additional, less, or alternate functionality, including that discussed elsewhere herein.

[0142] For instance, the at least one processor and/or associated transceiver may be programmed to: input the schedule-related data into a machine learning algorithm, model, module, or program trained to identify schedule conflict and/or resolutions to schedule conflicts based upon schedule-related data received from a senior and multiple caregivers.

[0143] The at least one processor and/or associated transceiver may be programmed to: when a schedule conflict is identified, analyze the schedule-related data to determine a resolution to the schedule conflict. The at least one processor and/or associated transceiver may be programmed to: when a schedule conflict is identified, input the schedule-related data into a machine learning algorithm, model, module, or program trained to identify resolutions to schedule conflicts based upon schedule-related data received from a senior and multiple caregivers.

[0144] The at least one processor and/or associated transceiver may be programmed to: invoke a calendar chatbot to verbally explain the schedule conflict and/or one or more proposed resolutions to the schedule conflict to the senior and/or one or more caregivers. The at least one processor and/or associated transceiver may be programmed to: receive verbal acceptance of the proposed resolution to the schedule conflict from the senior and/or one or more caregivers (such as via a mobile device or other computing device and a caregiver application). The at least one processor and/or associated transceiver may be programmed to: update the senior's calendar with a conflict resolution accepted by the senior or a caregiver, and/or generate and transmit electronic notifications of the scheduling change to the senior's mobile device and one or more caregiver mobile devices.

[0145] The at least one processor and/or associated transceiver may be programmed to: when a schedule conflict is identified, invoke a calendar chatbot to verbally engage the senior and/or one or more caregivers to verbally explain the schedule conflict and/or engage in conversation with the senior and/or one or more caregivers via a calendar chatbot to verbally arrive at a calendar or schedule conflict resolution.

[0146] In another aspect, a computer-implemented method for coordinating care and resolving schedule conflicts may be provided. The computer-implemented method may be performed by a care coordination support platform ("CCSP") computer system including at least one processor and/or associated transceiver in communication with at least one memory device. The computer-implemented method may include (1) building, creating, or otherwise generating a senior calendar (for a senior or other user) (i) with tasks, appointments, and/or activities (and/or other events) and (ii) with multiple caregivers having access to the senior calendar, wherein individual caregivers are assigned responsibilities for certain tasks, appointments, and/or activities; (2) collecting updated schedule information, mobile device data, sensor data (such as home, vehicle, or wearable sensor data), social media data, and/or other updated schedule-related data associated with the multiple caregivers and/or the senior (such as via wireless communication and/or data transmission over one or more radio frequency links); and/or (3) analyzing the updated schedule-related data to determine a schedule conflict to facilitate coordinating care for the senior. The method may include additional, less, or alternate actions, including those discussed elsewhere herein.

[0147] For instance, the method may include inputting the schedule-related data into a machine learning algorithm, model, module, or program trained to identify schedule conflict and/or resolutions to schedule conflicts based upon schedule-related data received from a senior and multiple caregivers.

[0148] The method may include, when a schedule conflict is identified, analyzing the schedule-related data to determine a resolution to the schedule conflict. The method may include, when a schedule conflict is identified, inputting the schedule-related data into a machine learning algorithm, model, module, or program trained to identify resolutions to schedule conflicts based upon schedule-related data received from a senior and multiple caregivers. The method may include invoking a calendar chatbot to verbally explain the schedule conflict and/or one or more proposed resolutions to the schedule conflict to the senior and/or one or more caregivers. The method may further include receiving verbal acceptance of the proposed resolution to the schedule conflict from the senior and/or one or more caregivers (such as via the senior's or caregiver's mobile device or other computing device and a care coordination application installed thereon).

[0149] The method may further include, when a schedule conflict is identified, invoking a calendar chatbot to verbally engage the senior and/or one or more caregivers to verbally explain the schedule conflict. The method may also include, when a schedule conflict is identified, engaging in conversation with the senior and/or one or more caregivers via a calendar chatbot to verbally arrive at a calendar or schedule conflict resolution. The method may further include updating the senior's calendar with a conflict resolution verbally or otherwise accepted by the senior or a caregiver, and/or generating and transmitting notifications of the scheduling change to the senior's mobile device and one or more caregiver mobile devices.

EXEMPLARY EVENT MONITORING & AUTOMATIC CALENDAR UPDATE

[0150] FIG. 22 illustrates an exemplary computer-implemented method for coordinating care and monitoring whether events have occurred as scheduled, or alternatively need to be rescheduled, via a care coordination support platform (“CCSP”) computer system **2200**. The CCSP computer system may include at least one processor and/or associated transceiver in communication with at least one memory device.

[0151] The computer-implemented method **2200** may include, via one or more processors and/or associated transceivers, building, creating, and/or otherwise generating **2202** an electronic calendar for a senior with multiple events (such as tasks, appointments, and/or activities, etc.), and with multiple caregivers (professional caregivers, service providers, medical offices, family members, friends, etc.) having electronic access to the calendar. The generating **2202** of the senior calendar may include the registering user and caregiver functionality; defining event (e.g., task, appointment, and/or activity) functionality; and/or assigning event functionality as discussed elsewhere herein.

[0152] The computer-implemented method **2200** may include, via one or more processors and/or associated transceivers, collecting **2204** GPS (Global Positioning System) and/or geofence location data associated with each event (e.g., task, appointment, and/or activity), and then add the GPS and geofence location data to the calendar data to create time and location data for each event (i.e., create event time and location data for each event).

[0153] Before and/or at the time of each event, the computer-implemented method **2200** may include, via one or more processors and/or associated transceivers, monitoring and/or collecting **2206** senior mobile device or wearable device GPS location and time data. For instance, the one or more processors and/or associated transceivers may query the mobile device, wearable device, or other computing devices associated with the senior (such as a smart home controller or smart vehicle computing device) to receive or otherwise determine the current GPS location of the senior.

[0154] The computer-implemented method **2200** may include, via one or more processors and/or associated transceivers, comparing **2208** the event location and time data with the senior's GPS location and time data to facilitate determining whether the event took place as scheduled, or if the

event did not occur as scheduled, and needs to be rescheduled.

[0155] If the event location matches the senior's GPS location before and/or at the time of the event, the computer-implemented method **2200** may include, via one or more processors and/or associated transceivers, updating **2210** the electronic calendar of the senior to indicate that the event (task, activity, appointment, etc.) has been completed. After which, an electronic notification indicating event completion may be transmitted to the mobile devices or other computing devices of one or more caregivers and/or the senior.

[0156] If the event location does not match the senior's GPS location before and/or at the time of the event, the computer-implemented method **2200** may include, via one or more processors and/or associated transceivers, generating and transmitting **2212** electronic notifications indicating that the event did not occur as scheduled, and/or that the event needs to be rescheduled, to the mobile devices or other computing devices of the senior and/or one or more caregivers. The computer-implemented method **2200** may then include, via one or more processors and/or associated transceivers, triggering **2214** the chat to engage the senior and/or one or more caregivers to reschedule the event.

[0157] In some embodiments, the computer system and method may classify each event as being either critical or non-critical events. For instance, taking prescribed medications at the prescribed time, and/or attending medical appointments may be deemed to be critical events that should be rescheduled if they are not completed as originally scheduled. Other events, such as attending a playing or seeing a movie may be deemed non-critical, and may or may not need to be rescheduled.

[0158] In some embodiments, the one or more processors may analyze updated schedule-related information of the senior and/or caregivers to identify potential replacement times for the event, and the chatbot may then verbally propose one or more replacement times and/or locations to the senior. Additionally or alternatively, the chatbot may verbally engage the senior and/or caregiver, and ask for one or more replacement times and/or locations for the event. The method **2200** may include additional, less, or alternate actions, including those discussed elsewhere herein.

[0159] In one aspect, a care coordination support platform ("CCSP") computer system for coordinating care and monitoring whether scheduled calendar events have or have not occurred as scheduled may be provided. The CCSP computer system may include at least one processor and/or associated transceiver in communication with at least one memory device. The at least one processor and/or associated transceiver may be programmed to: (1) build, create, or otherwise generate a senior calendar (for a senior or other user) (i) with events (such as tasks, appointments, and/or activities) and (ii) with multiple caregivers having access to the senior calendar, wherein individual caregivers are assigned responsibilities for certain tasks, appointments, and/or activities; (2) collect GPS (Global Positioning System) and/or geofence location data associated with each scheduled event; (3) add the location data associated with each event to calendar (or time) data to create time and location data for each event; (4) before and/or at the time of each event, monitor and/or collect senior GPS location and time data defining a location of the senior at a specific time, the senior GPS location data originating from a mobile device, wearable, vehicle controller, smart home controller, or other computing device associated with the senior; and/or (5) compare the event location and time data with the senior GPS location and time data to facilitate monitoring event completion and care coordination. The CCSP computer system may include additional, less, or alternate functionality, including that discussed elsewhere herein.

[0160] For instance, the at least one processor and/or associated transceiver may be programmed to: if the event location and time data matches the senior GPS location and time data, update the electronic calendar of the senior to indicate that the event was completed.

[0161] The at least one processor and/or associated transceiver may be programmed to: if the event location and time data does not match the senior GPS location and time data, update the electronic calendar of the senior to indicate that the event was not completed, and/or generate and transmit an electronic notification to the mobile device or other computing device of the senior and/or one or

more caregivers indicating that the event was not completed as scheduled.

[0162] The at least one processor and/or associated transceiver may be programmed to: if the event location and time data does not match the senior GPS location and time data, trigger or invoke a calendar chatbot to verbally engage the senior and/or one or more caregivers in conversation to reschedule the event or otherwise resolve an uncompleted event.

[0163] In another aspect, a computer-implemented method for coordinating care and monitoring whether scheduled calendar events have or have not occurred as scheduled. The computer-implemented method may be performed by a care coordination support platform (“CCSP”) computer system including at least one processor and/or associated transceiver in communication with at least one memory device. The computer-implemented method may include (1) building, creating, or generating a senior calendar (for a senior or other user) (i) with events (such as tasks, appointments, and/or activities) and (ii) with multiple caregivers having access to the senior calendar, wherein individual caregivers are assigned responsibilities for certain tasks, appointments, and/or activities; (2) collecting GPS (Global Positioning System) and/or geofence location data associated with each scheduled event; (3) adding the location data associated with each event to calendar (or time) data to create time and location data for each event; (4) before and/or at the time of each event, monitoring and/or collecting senior GPS location and time data defining a location of the senior at a specific time, the senior GPS location data originating from a mobile device, wearable, vehicle controller, smart home controller, or other computing device associated with the senior; and/or (5) comparing the event location and time data with the senior GPS location and time data to facilitate monitoring event completion and care coordination.

[0164] The method may include, if the event location and time data matches the senior GPS location and time data, updating the electronic calendar of the senior to indicate that the event was completed. The method may include, if the event location and time data does not match the senior GPS location and time data, updating the electronic calendar of the senior to indicate that the event was not completed, and/or generating and transmitting an electronic notification to the mobile device or other computing device of the senior and/or one or more caregivers indicating that the event was not completed.

[0165] The method may include, if the event location and time data does not match the senior GPS location and time data, triggering or invoking a calendar chatbot to verbally engage the senior and/or one or more caregivers in conversation to reschedule the event or otherwise resolve an uncompleted event.

MACHINE LEARNING & OTHER MATTERS

[0166] The computer systems and computer-implemented methods discussed herein may include additional, less, or alternate actions and/or functionalities, including those discussed elsewhere herein. The computer systems may include or be implemented via computer-executable instructions stored on non-transitory computer-readable media. The methods may be implemented via one or more local or remote processors, transceivers, servers, and/or sensors (such as processors, transceivers, servers, and/or sensors mounted on mobile computing devices, or associated with smart infrastructure or remote servers), and/or via computer executable instructions stored on non-transitory computer-readable media or medium.

[0167] In some embodiments, a care coordination support platform computing device is configured to implement machine learning, such that the care coordination support platform computing device “learns” to analyze, organize, and/or process data without being explicitly programmed. Machine learning may be implemented through machine learning methods and algorithms (“ML methods and algorithms”). In an exemplary embodiment, a machine learning module (“ML module”) is configured to implement ML methods and algorithms. In some embodiments, ML methods and algorithms are applied to data inputs and generate machine learning outputs (“ML outputs”). Data inputs may include but are not limited to: user data, caregiver data, sensor data, assignment data, calendar data, task data, and/or alert data. ML outputs may include but are not limited to: user data,

caregiver data, calendar data, task data, and/or assignment data. In some embodiments, data inputs may include certain ML outputs.

[0168] In some embodiments, at least one of a plurality of ML methods and algorithms may be applied, which may include but are not limited to: linear or logistic regression, instance-based algorithms, regularization algorithms, decision trees, Bayesian networks, cluster analysis, association rule learning, artificial neural networks, deep learning, combined learning, reinforced learning, dimensionality reduction, and support vector machines. In various embodiments, the implemented ML methods and algorithms are directed toward at least one of a plurality of categorizations of machine learning, such as supervised learning, unsupervised learning, and reinforcement learning.

[0169] In one embodiment, the ML module employs supervised learning, which involves identifying patterns in existing data to make predictions about subsequently received data. Specifically, the ML module is “trained” using training data, which includes example inputs and associated example outputs. Based upon the training data, the ML module may generate a predictive function which maps outputs to inputs and may utilize the predictive function to generate ML outputs based upon data inputs. The example inputs and example outputs of the training data may include any of the data inputs or ML outputs described above. For example, a ML module may receive training data comprising user data, caregiver data, and assignment data associated with the user data and caregiver data. The ML module may then generate a model which maps assignment data to aspects of user data and caregiver data. The ML module may then generate assignment data as a ML output based upon subsequently received user data and caregiver data.

[0170] In another embodiment, a ML module may employ unsupervised learning, which involves finding meaningful relationships in unorganized data. Unlike supervised learning, unsupervised learning does not involve user-initiated training based upon example inputs with associated outputs. Rather, in unsupervised learning, the ML module may organize unlabeled data according to a relationship determined by at least one ML method/algorithm employed by the ML module. Unorganized data may include any combination of data inputs and/or ML outputs as described above. For example, a ML module may receive unlabeled data comprising user data, caregiver data, and calendar data. The ML module may employ an unsupervised learning method such as “clustering” to identify patterns and organize the unlabeled data into meaningful groups. The newly organized data may be used, for example, to generate a model which associates user data and caregiver data to calendar data.

[0171] In yet another embodiment, a ML module may employ reinforcement learning, which involves optimizing outputs based upon feedback from a reward signal. Specifically, the ML module may receive a user-defined reward signal definition, receive a data input, utilize a decision-making model to generate a ML output based upon the data input, receive a reward signal based upon the reward signal definition and the ML output, and alter the decision-making model so as to receive a stronger reward signal for subsequently generated ML outputs. Other types of machine learning may also be employed, including deep or combined learning techniques.

[0172] The reward signal definition may be based upon any of the data inputs or ML outputs described above. For example, a ML module may implement reinforcement learning in generating assignment data for caregivers. The ML module may utilize a decision-making model to generate assignment data for caregivers based upon task data, and may further receive user-satisfaction data indicating a level of satisfaction experienced by a user and a caregiver who engaged in a transaction (e.g., the caregiver carrying out a task for the user). A reward signal may be generated by comparing the user-satisfaction data to an assignment score between the user and the caregiver.

[0173] Based upon the reward signal, the ML module may update the decision-making model such that subsequently generated assignment scores more accurately predict user satisfaction. For example, the ML module may determine that a specific caregiver has taken the user to four doctor's appointments. The user may enjoy the caregiver taking the user to the doctor's appointments, and

the caregiver may enjoy taking the user to the doctor's appointments because the doctor's appointments may be close to the caregiver's house. Therefore, the user and the caregiver may both rate the "transaction" highly. Accordingly, the ML module may learn to automatically assign doctor's appointments to the specific caregiver.

ADDITIONAL CONSIDERATIONS

[0174] With the foregoing, users and caregivers may opt-in or register to a care coordination support platform program or other type of program. After the users and caregivers give their affirmative consent or permission, a care coordination support platform remote server may collect data from the mobile devices, user computing devices, smart home controllers, smart vehicles, autonomous or semi-autonomous vehicles, smart infrastructure, smart buildings, smart aerial devices (e.g., drones), and/or other smart devices, such as with the permission or affirmative consent of the users and caregivers. The data collected may be related to user activities and/or user/caregiver schedules and current locations.

[0175] As will be appreciated based upon the foregoing specification, the above-described embodiments of the disclosure may be implemented using computer programming or engineering techniques including computer software, firmware, hardware or any combination or subset thereof. Any such resulting program, having computer-readable code means, may be embodied or provided within one or more computer-readable media, thereby making a computer program product, i.e., an article of manufacture, according to the discussed embodiments of the disclosure. The computer-readable media may be, for example, but is not limited to, a fixed (hard) drive, diskette, optical disk, magnetic tape, semiconductor memory such as read-only memory (ROM), and/or any transmitting/receiving medium such as the Internet or other communication network or link. The article of manufacture containing the computer code may be made and/or used by executing the code directly from one medium, by copying the code from one medium to another medium, or by transmitting the code over a network.

[0176] These computer programs (also known as programs, software, software applications, "apps", or code) include machine instructions for a programmable processor, and can be implemented in a high-level procedural and/or object-oriented programming language, and/or in assembly/machine language. As used herein, the terms "machine-readable medium" "computer-readable medium" refers to any computer program product, apparatus and/or device (e.g., magnetic discs, optical disks, memory, Programmable Logic Devices (PLDs)) used to provide machine instructions and/or data to a programmable processor, including a machine-readable medium that receives machine instructions as a machine-readable signal. The "machine-readable medium" and "computer-readable medium," however, do not include transitory signals. The term "machine-readable signal" refers to any signal used to provide machine instructions and/or data to a programmable processor.

[0177] As used herein, a processor may include any programmable system including systems using micro-controllers, reduced instruction set circuits (RISC), application specific integrated circuits (ASICs), logic circuits, and any other circuit or processor capable of executing the functions described herein. The above examples are example only, and are thus not intended to limit in any way the definition and/or meaning of the term "processor."

[0178] As used herein, the terms "software" and "firmware" are interchangeable, and include any computer program stored in memory for execution by a processor, including RAM memory, ROM memory, EPROM memory, EEPROM memory, and non-volatile RAM (NVRAM) memory. The above memory types are example only, and are thus not limiting as to the types of memory usable for storage of a computer program.

[0179] In one embodiment, a computer program is provided, and the program is embodied on a computer readable medium. In an exemplary embodiment, the system is executed on a single computer system, without requiring a connection to a sever computer. In a further embodiment, the system is being run in a Windows® environment (Windows is a registered trademark of Microsoft

Corporation, Redmond, Washington). In yet another embodiment, the system is run on a mainframe environment and a UNIX® server environment (UNIX is a registered trademark of X/Open Company Limited located in Reading, Berkshire, United Kingdom). The application is flexible and designed to run in various different environments without compromising any major functionality. In some embodiments, the system includes multiple components distributed among a plurality of computing devices. One or more components may be in the form of computer-executable instructions embodied in a computer-readable medium. The systems and processes are not limited to the specific embodiments described herein. In addition, components of each system and each process can be practiced independent and separate from other components and processes described herein. Each component and process can also be used in combination with other assembly packages and processes.

[0180] In some embodiments, registration of users for the care coordination support platform includes opt-in informed consent of users to data usage by the smart home devices, wearable devices, mobile devices, autonomous vehicles, and/or smart vehicles consistent with consumer protection laws and privacy regulations. In some embodiments, the user data, the caregiver data, and/or other collected data may be anonymized and/or aggregated prior to receipt such that no personally identifiable information (PII) is received. In other embodiments, the system may be configured to receive user and caregiver data and/or other collected data that is not yet anonymized and/or aggregated, and thus may be configured to anonymize and aggregate the data. In such embodiments, any PII received by the system is received and processed in an encrypted format, or is received with the consent of the individual with which the PII is associated. In situations in which the systems discussed herein collect personal information about individuals, or may make use of such personal information, the individuals may be provided with an opportunity to control whether such information is collected or to control whether and/or how such information is used. In addition, certain data may be processed in one or more ways before it is stored or used, so that personally identifiable information is removed.

[0181] As used herein, an element or step recited in the singular and proceeded with the word “a” or “an” should be understood as not excluding plural elements or steps, unless such exclusion is explicitly recited. Furthermore, references to “exemplary embodiment” or “one embodiment” of the present disclosure are not intended to be interpreted as excluding the existence of additional embodiments that also incorporate the recited features.

[0182] The patent claims at the end of this document are not intended to be construed under 35 U.S.C. § 112(f) unless traditional means-plus-function language is expressly recited, such as “means for” or “step for” language being expressly recited in the claim(s).

[0183] This written description uses examples to disclose the disclosure, including the best mode, and also to enable any person skilled in the art to practice the disclosure, including making and using any devices or systems and performing any incorporated methods. The patentable scope of the disclosure is defined by the claims, and may include other examples that occur to those skilled in the art. Such other examples are intended to be within the scope of the claims if they have structural elements that do not differ from the literal language of the claims, or if they include equivalent structural elements with insubstantial differences from the literal language of the claims.

Claims

1-27. (canceled)

28. A computer system for continuously monitoring one or more audible inputs performed on a remote computing device, the computer system comprising at least one processor, at least one memory device in communication with the at least one processor, and a chatbot in communication with the at least one processor, the at least one processor configured to: store, in the at least one memory device, an electronic care schedule for a user including a plurality of scheduled events,

wherein one or more of the plurality of scheduled events are assigned to at least one caregiver of a plurality of caregivers; train a machine learning model using training data including (i) user data associated with the user and (ii) caregiver data associated with the plurality of caregivers, and (iii) the electronic care schedule; receive, via the chatbot and at least one microphone of a first client device, a first audible input defining a first new event associated with the user; determine whether a schedule conflict exists based upon the first new event and at least one conflicting scheduled event included in the electronic care schedule; if the determination is that the schedule conflict exists, the at least one processor is further configured to: output, from the trained machine learning model by applying the first new event into the trained machine learning model, assignment data indicating at least one resolution to the schedule conflict, wherein outputting the assignment data includes identifying at least one of (i) one or more replacement caregivers for the first new event or the at least one conflicting scheduled event or (ii) one or more replacement times for the first new event or the at least one conflicting scheduled event; and generate and audibly present, via the chatbot and at least one speaker of the first client device, a first audible output advising that the schedule conflict existed and providing the at least one resolution; if the determination is that the schedule conflict does not exist, the at least one processor is further configured to: generate and audibly present, via the chatbot and the at least one speaker of the first client device, a second audible output requesting an audible response from the user confirming that the user desires to add the first new event to the electronic care schedule; determine, based on at least one of sensor data or location data received from one or more computing devices associated with the user, whether the first new event has been completed; if the determination is that the first new event has been completed, (i) automatically check a box indicating that the first new event has been completed, (ii) cause display of the box on the first client device, and (iii) generate an assignment score associated with the first new event; and if the determination is that the first new event has not been completed, (i) generate an alert indicating that the first new event has yet to be completed, and (ii) cause display of the alert on at least one of the first client device or a second client device associated with one of the plurality of caregivers; and re-train the trained machine learning model by inputting additional user data, additional caregiver data, and the assignment score.

29. The computer system of claim 28, wherein the at least one processor is further configured to register the user via a computer application executing on the first client device, wherein registering the user includes receiving user personal and scheduling data including the plurality of scheduled events inputted by the user into the computer application.

30. The computer system of claim 28, wherein the at least one processor is further configured to register a caregiver of the plurality of caregivers via a computer application executing on a second client device, wherein registering the caregiver of the plurality of caregivers includes receiving caregiver personal and scheduling data including the corresponding caregiver's scheduled events and the scheduled events of the user assigned to the corresponding caregiver inputted by the caregiver into the computer application.

31. The computer system of claim 28, wherein when the audible response is a confirmation to add the first new event, the at least one processor is further configured to: electronically add the first new event to the electronic care schedule; electronically assign the first new event to a first caregiver of the plurality of caregivers based upon caregiver personal and scheduling data associated with the first caregiver; perform a lookup within the one or more memory device to determine that a second client device is registered with the first caregiver; and electronically transmit, to the second client device, computer instructions to cause the second client device to display, via a computer application executing on the second client device and a display screen of the second client device, an update to the electronic care schedule to the first caregiver.

32. The computer system of claim 28, wherein the at least one processor is further configured to: receive, via the chatbot, a converted text question of an audible question associated with the first new event inputted using at least one of the first client device or a second client device associated

with a first caregiver of the plurality of caregivers, wherein the audible question is a natural-language voice question inputted into the chatbot; convert the text question into a search query; execute the search query against an event database stored in the at least one memory device; and cause, via the chatbot to at least one of the first client device or the second client device, an audible answer to the audible question to be presented based upon the search query returning the first new event.

33. The computer system of claim 28, wherein the at least one processor is further configured to: receive sensor data from sensors associated with at least one of the user or a first caregiver of the plurality of caregivers, wherein the sensors include at least one of mobile device sensors, vehicle telematics sensors, home telematics sensors, or smart home device sensors; determine, based upon the received sensor data, whether the first new event has been carried out by the first caregiver; and transmit a notification to a second client device associated with at least one caregiver of the plurality of caregivers, wherein the notification indicates whether the first new event has been carried out by the first caregiver.

34. The computer system of claim 28, wherein the at least one processor is further configured to: update, via a computer application executing on a second client device associated with a first caregiver, an electronic calendar of the first caregiver in response to at least one new event being assigned to the first caregiver.

35. The computer system of claim 28, wherein the at least one processor is further configured to: monitor an active time during which each of the plurality of caregivers cares for the user; store the monitored active time for each caregiver; compare the active time for each caregiver with the active time for each other caregiver; and display, on a client device, the compared active time for each caregiver, wherein the compared active time is displayed for a time interval, and wherein the time interval is at least one of a week, two weeks, three weeks, a month, or a year.

36. The computer system of claim 28, wherein the at least one processor is further configured to: receive medication data associated with the user from at least one of the user or the plurality of caregivers; provide, via a computer application executing on at least one of the first client device or a second client device associated with one or more of the plurality of caregivers, a daily medication list; and enable, via the computer application, input from at least one of the user and the one or more caregivers, wherein the input indicates when the user takes each medication from the daily medication list.

37. A computer-implemented method for continuously monitoring one or more audible inputs performed on a remote computing device, the computer-implemented method performed by a computer system including at least one processor, at least one memory device in communication with the at least one processor, and a chatbot in communication with the at least one processor, the computer-implemented method comprising: storing, in the at least one memory device, an electronic care schedule for a user including a plurality of scheduled events, wherein one or more of the plurality of scheduled events are assigned to at least one caregiver of a plurality of caregivers; training a machine learning model using training data including (i) user data associated with the user and (ii) caregiver data associated with the plurality of caregivers, and (iii) the electronic care schedule; receiving, via the chatbot and at least one microphone of a first client device, a first audible input defining a first new event associated with the user; determining whether a schedule conflict exists based upon the first new event and at least one conflicting scheduled event included in the electronic care schedule; if the determination is that the schedule conflict exists, the computer-implemented method further comprises: outputting, from the trained machine learning model by applying the first new event into the trained machine learning model, assignment data indicating at least one resolution to the schedule conflict outputting the assignment data comprises identifying at least one of (i) one or more replacement caregivers for the first new event or the at least one conflicting scheduled event or (ii) one or more replacement times for the first new event or the at least one conflicting scheduled event; and generating and audibly

presenting, via the chatbot and at least one speaker of the first client device, a first audible output advising that the schedule conflict existed and providing the at least one resolution; if the determination is that the schedule conflict does not exist, the computer-implemented method further comprises: generating and audibly presenting, via the chatbot and the at least one speaker of the first client device, a second audible output requesting an audible response from the user confirming that the user desires to add the first new event to the electronic care schedule; determine, based on at least one of sensor data or location data received from one or more computing devices associated with the user, whether the first new event has been completed; if the determination is that the first new event has been completed, (i) automatically check a box indicating that the first new event has been completed, (ii) cause display of the box on the first client device, and (iii) generate an assignment score associated with the first new event; and if the determination is that the first new event has not been completed, (i) generate an alert indicating that the first new event has yet to be completed, and (ii) cause display of the alert on at least one of the first client device or a second client device associated with one of the plurality of caregivers; and re-training the trained machine learning model by inputting additional user data, additional caregiver data, and the assignment score.

38. The computer-implemented method of claim 37 further comprising registering the user via a computer application executing on the first client device, wherein registering the user comprises receiving user personal and scheduling data including the plurality of scheduled events inputted by the user into the computer application.

39. The computer-implemented method of claim 37 further comprising registering a caregiver of the plurality of caregivers via a computer application executing on a second client device, wherein registering the caregiver of the plurality of caregivers comprises receiving caregiver personal and scheduling data including the corresponding caregiver's scheduled events and the scheduled events of the user assigned to the corresponding caregiver inputted by the caregiver into the computer application.

40. The computer-implemented method of claim 37, wherein when the audible response is a confirmation to add the first new event, the computer-implemented method further comprises: electronically adding the first new event to the electronic care schedule; electronically assigning the first new event to a first caregiver of the plurality of caregivers based upon caregiver personal and scheduling data associated with the first caregiver; performing a lookup within the one or more memory device to determine that a second client device is registered with the first caregiver; and electronically transmitting, to the second client device, computer instructions to cause the second client device to display, via a computer application executing on the second client device and a display screen of the second client device, an update to the electronic care schedule to the first caregiver.

41. The computer-implemented method of claim 37 further comprising: receiving, via the chatbot, a converted text question of an audible question associated with the first new event inputted using at least one of the first client device or a second client device associated with a first caregiver of the plurality of caregivers, wherein the audible question is a natural-language voice question inputted into the chatbot; converting the text question into a search query; executing the search query against an event database stored in the at least one memory device; and causing, via the chatbot to at least one of the first client device or the second client device, an audible answer to the audible question to be presented based upon the search query returning the first new event.

42. The computer-implemented method of claim 37 further comprising: receiving sensor data from sensors associated with at least one of the user or a first caregiver of the plurality of caregivers, wherein the sensors include at least one of mobile device sensors, vehicle telematics sensors, home telematics sensors, or smart home device sensors; determining, based upon the received sensor data, whether the first new event has been carried out by the first caregiver; and transmitting a notification to a second client device associated with at least one caregiver of the plurality of

caregivers, wherein the notification indicates whether the first new event has been carried out by the first caregiver.

43. The computer-implemented method of claim 37 further comprising: updating, via a computer application executing on a second client device associated with a first caregiver, an electronic calendar of the first caregiver in response to at least one new event being assigned to the first caregiver.

44. The computer-implemented method of claim 37 further comprising: monitoring an active time during which each of the plurality of caregivers cares for the user; storing the monitored active time for each caregiver; comparing the active time for each caregiver with the active time for each other caregiver; and displaying, on a client device, the compared active time for each caregiver, wherein the compared active time is displayed for a time interval, and wherein the time interval is at least one of a week, two weeks, three weeks, a month, or a year.

45. The computer-implemented method of claim 37 further comprising: receiving medication data associated with the user from at least one of the user or the plurality of caregivers; providing, via a computer application executing on at least one of the first client device or a second client device associated with one or more of the plurality of caregivers, a daily medication list; and enabling, via the computer application, input from at least one of the user and the one or more caregivers, wherein the input indicates when the user takes each medication from the daily medication list.

46. A non-transitory computer-readable medium having computer-executable instructions embodied thereon, wherein when executed by computer system including at least one processor, at least one memory device in communication with the at least one processor, and a chatbot in communication with the at least one processor, the computer-executable instructions cause the at least one processor to: store, in the at least one memory device, an electronic care schedule for a user including a plurality of scheduled events, wherein one or more of the plurality of scheduled events are assigned to at least one caregiver of a plurality of caregivers; train a machine learning model using training data including (i) user data associated with the user and (ii) caregiver data associated with the plurality of caregivers, and (iii) the electronic care schedule; receive, via the chatbot and at least one microphone of a first client device, a first audible input defining a first new event associated with the user; determine whether a schedule conflict exists based upon the first new event and at least one conflicting scheduled event included in the electronic care schedule; if the determination is that the schedule conflict exists, the at least one processor is further configured to: output, from the trained machine learning model by applying the first new event into the trained machine learning model, assignment data indicating at least one resolution to the schedule conflict wherein outputting the assignment data includes identifying at least one of (i) one or more replacement caregivers for the first new event or the at least one conflicting scheduled event or (ii) one or more replacement times for the first new event or the at least one conflicting scheduled event; and generate and audibly present, via the chatbot and at least one speaker of the first client device, a first audible output advising that the schedule conflict existed and providing the at least one resolution; if the determination is that the schedule conflict does not exist, the at least one processor is further configured to: generate and audibly present, via the chatbot and the at least one speaker of the first client device, a second audible output requesting an audible response from the user confirming that the user desires to add the first new event to the electronic care schedule; determine, based on at least one of sensor data or location data received from one or more computing devices associated with the user, whether the first new event has been completed; if the determination is that the first new event has been completed, (i) automatically check a box indicating that the first new event has been completed, and (ii) cause display of the box on the first client device, and (iii) generate an assignment score associated with the first new event; and if the determination is that the first new event has not been completed, (i) generate an alert indicating that the first new event has yet to be completed, (ii) cause display of the alert on at least one of the first client device or a second client device associated with one of the plurality of caregivers; and re-

train the trained machine learning model by inputting additional user data, additional caregiver data, and the assignment score.

47. The non-transitory computer-readable medium of claim 46, wherein when the audible response is a confirmation to add the first new event, the computer-executable instructions further cause the at least one processor to: electronically add the first new event to the electronic care schedule; electronically assign the first new event to a first caregiver of the plurality of caregivers based upon caregiver personal and scheduling data associated with the first caregiver; perform a lookup within the one or more memory device to determine that a second client device is registered with the first caregiver; and electronically transmit, to the second client device, computer instructions to cause the second client device to display, via a computer application executing on the second client device and a display screen of the second client device, an update to the electronic care schedule to the first caregiver.
